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Models of Bond Prices

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Why bonds?

We study bond prices in detail for three related reasons.

1. The most obvious is that bonds – and the broader class of so-called fixed-income assets and their derivatives – are the largest asset class in the world.[1]
2. A more subtle reason is that embedded somewhere in the valuation of every equity asset, every corporate capital budgeting decision, every home mortgage, every exchange-rate transaction, and every government budget decision, is a bond of one form or another. Bonds are fundamental building blocks for every other financial valuation you are likely to encounter.[2]
3. The reason bonds are so important for our current purpose, that is, understanding the foundations of asset-pricing, is that the simplest form of a bond – something we define shortly and call a *default-free, pure-discount bond* – is the most direct measurement of the pricing kernel that we have.[3] We will see that *all* of the properties of these bond prices are driven *entirely* by the properties of the pricing kernel.

[1] In the US, bond markets are about twice the size of the stock market, with about 4 times the daily trading volume. An amount roughly equal to US *annual* GDP is traded every 10 *days*. And the relative magnitudes are similar at a global scale.

[2] The first concept you learn in finance is *present value*, which is just a special case of bond pricing.

[3] Embedded in every other variety of bond is a default-free, pure-discount bond. If bonds are fundamental building blocks for financial valuations, default-free, pure-discount bonds are the fundamental building blocks for bond valuations.

How will theory be connected to data?

We begin by defining bond prices and related objects, then describe their connection with the pricing kernel in arbitrage-free economies. Of particular interest is the *yield curve* – sometimes called the *term structure of interest rates* – which is the graph of pure-discount bond prices for bonds with different time horizons, which we call *maturities*, plotted against their maturities.[4] We find it more convenient to first transform prices into yields which have the same basic units as the pricing kernel, hence the term *yield curve*. But don't be misled by the terminology: yields are just a convenient way to quote prices. Roughly speaking, the dynamics of the pricing kernel in our theoretical model are revealed to us through the shape of the average yield curve, the persistence of changes in yields, and the volatility of yields. And since we have lots of data on the yield curve, we will have lots of indirect information about the pricing kernel too.

[4] In the financial press you'll see lots of different plots casually described as a *yield curve*. Often these involve bonds with extra cashflows, imbedded options, and a variety of other complications. Don't be misled. The *yield curve* we find most useful has a precise definition which we discuss below.

What's the pricing kernel again?

Formally, we apply the no-arbitrage theorem in a dynamic setting. Consider a Markov environment with state variable z_t. The no-arbitrage theorem states that there exists a positive pricing kernel m that satisfies

$$E\left[m(z_t, z_{t+1})r(z_t, z_{t+1}) \mid z_t\right] \quad = \quad 1,$$

which we express in shorthand as

$$E_t\left[m_{t+1}r_{t+1}\right] \quad = \quad 1 \tag{1}$$

for the gross return on any asset, r_{t+1}. Note that the left-hand-side of this expression is a function of z_t, the state of the economy at date t, which in equilibrium must be equal a constant value of 1, for all values of z_t, and all time periods. In other words, don't be deceived by the compact form of the notation. This one equation doesn't represent one restriction, it represents a very large set of restrictions – one for each possible value of z_t – for a very large set of assets, since it applies to *any and all* returns, r_{t+1}.

We show how this works for bonds of different maturities and use observed properties of interest rates (means, standard deviations, autocorrelations) to infer parameter values for some popular models. Our examples are loglinear. In math jargon they're called *exponential-affine* models, or sometimes just *affine* models. They have the advantages of tractability and computational simplicity: the solutions sometimes look complicated, but they are clear enough that we can make some intuitive sense of them, and computing solutions to numerical versions of the model is trivial.

Why does it feel like we're working backwards?

You will notice a different methodology than we used earlier. Our usual and customary approach has been to start with a model of preferences and investment opportunities, then build up a pricing kernel from the two basic economic foundations of optimization and equilibrium. We would then compare the predictions of that model to observed properties of data to see whether that fundamental model of the pricing kernel was useful for understanding that data.

We're now going to work in reverse. We start by using equation (1) to justify the existence of a stochastic process, m_{t+1}. Then we will show how that stochastic process for the pricing kernel is related to the stochastic process for bond prices of different maturities. Then

we will look at properties of bond-price data to infer the most useful stochastic process for m_{t+1}. We call that *reverse engineering* the pricing kernel. For a wide range of real-world applications, the modeling exercise can stop there: it's enough to know the process for the pricing kernel without know where it comes from. But for us, the last step will compare the properties of reverse-engineered pricing kernels to theoretical pricing kernels that we build up from first principles, thereby ending up at our old starting point.

Definitions: bond prices, yields, and forward rates

The simplest bond.

As always, we start with the simplest possible example. Only after gaining a thorough understanding of the simplest example, can we then build in more complexity. As mentioned earlier, the simplest possible bond is a default-free[5], pure-discount bond. But what does that terminology mean?

[5] Who is immune from default? In practice it'll be a stable and solvent government agency, like the US Treasury.

Since an asset is fined by the process that generates its cashflows, we begin by defining the cashflows of this simplest of bonds. An investor who buys this bond makes a payment at date t that we naturally call the *bond price*.[6] They then receives a payment of exactly 1 with perfect certainty in period $t+n$. That's it! There are no cashflows before or after $t+n$, and there is no risk of getting paid anything other than 1. Could anything be simpler? We call the length of time, n, that the investor must wait to collect this single payment the *maturity* of the bond.[7]

[6] We sometimes call this a *long* position. The agent who sells the bond has a *short* position, for which the basic cashflows have the opposite signs.

[7] Sometimes n is called the *term* of the bond, which is the origin of jargon *term structure*.

We denote the price of such a bond by $q_t^{(n)}$, a natural generalization of the notation for the price of a one-period $q_t^{(1)}$ we've already been using.[8]

[8] The jargon should now be clear. We say *default-free* because the payment of 1 is guaranteed and involves no risk. We say *pure discount* because there are no cashflows besides the single payment of 1 at maturity, hence, the price is simply how much we *discount* that 1. Another common name for this bond with just one single payment at maturity is a *zero-coupon* bond.

Yields. We express a bond price in a more user-friendly form as a *yield to maturity*, or simply yield. Our traditional definition of a return, *i.e.*, future payoff divided by current price, for a maturity-n pure discount bond is $1/q_t^{(n)}$. But that's not a one-period return, it's a return that I have to wait n periods to collect. The thing we're most interested in is the pricing kernel and it's a one-period object. Therefore, it's customary to calculate an average per-period return for a multi-period bond so that the units are the same one-period units as the pricing kernel, the interest rate, *etc*. In other words, we want to think about a constant rate of return y, so that if we invest an amount q and

earn a constant return of $(1+y)$ each period for n periods, we will end up with 1:

$$q\underbrace{(1+y)(1+y)\cdots(1+y)}_{\text{n times}} \quad = \quad 1 \quad \Rightarrow y = q^{-1/n} - 1.$$

But there's an implicit arbitrary assumption in this calculation: the return is paid only at the very end of each time period. This might seem natural, but since we're just inventing the definition for y, can't we adopt virtually *any* assumption about when this fictional interest is paid? What if we assumed that 1/4-th of y is the rate of return paid every 1/4-th of a time period for n periods? Adopting this convention results in a different value of the yield, y, even though the size of the investment, q, and the size of the total payoff in n periods, 1, stay the same:

$$q\underbrace{(1+y/4)(1+y/4)\cdots(1+y/4)}_{\text{4n times}} \quad = \quad 1 \quad \Rightarrow y = 4[q^{-1/(4n)} - 1].$$

But what's so special about dividing each period into quarters? Nothing! We could've just as easily divided it into an arbitrary number of periods, say k, and come up with a different – and equally arbitrary – definition for the yield:

$$q\underbrace{(1+y/k)(1+y/k)\cdots(1+y/k)}_{\text{kn times}} \quad = \quad 1 \quad \Rightarrow y = k[q^{-1/(kn)} - 1].$$

In other words, every arbitrary choice for k provides another arbitrary definition for y.[9] When n is measured in years, you'll often see yields arbitrarily quoted with compounding defined over a fraction of a year like "semi-annual compounded, which is $k = 2$, or"quarterly compounding", which is $k = 4$, or "monthly compounding", which is $k = 12$. For reasons that will become apparent as we begin to develop a theory of bond pricing, we'll find it convenient use an arbitrary definition of a yield referred to as "continuous compounding". But remember, the choice is a matter of convenience, not substance.[10]

[9] This can be so confusing to the finance-illiterate average person that governments often create consumer-protection regulations stipulating how interest-rate compounding must be quoted for consumer loans.

[10] And don't forget that we're working with yields rather than prices strictly as a matter of convenience, not substance.

Consider the compounding of interest over every infinitesimally small instant of time, which would imply an arbitrarily large number of periods of compounding at an arbitrarily small rate of return. Luckily, we know a bit of math that we can apply this. Simply take the formula for compounding over k periods and take the limit as

$k \to \infty$:

$$\begin{aligned} \Rightarrow y &= \lim_{k\to\infty} k[q^{-1/(kn)} - 1] \\ &= \lim_{h\to 0}\left[\frac{e^{h(-1/n)\log q} - 1}{h}\right] \\ &= (-1/n)\log q, \end{aligned}$$

where the second equation replaces k with $1/h$ and the limit $k \to \infty$ with the equivalent limit $h \to 0$. The last equality comes via application of our old friend from first-year calculus, l'Hopital's rule.[11]

[11] Recall this is a clever way to solve what initially looks like a 0-divided-by-0 problem. Consider two functions, say $f(x)$ and $g(x)$, for which $f(0) = 0$ and $g(0) = 0$. The limit as $x \to 0$ of their ratio, $f(h)/g(h)$ is equal to the ratio of their derivatives, $f'(0)/g'(0)$.

The convenience here is the loglinearity of the yield in terms of the bond price. Note that no information is lost through this transformation: just as we can convert prices to yields, we can also always convert yields to prices, $q = \exp\{-ny\}$. Also note that since we will want to track the date of a price as well as its maturity, we will attach that information to our notation for yields just as we do for prices, and write them as

$$y_t^{(n)} = (-1/n)\log q_t^{(n)}.$$

Digression on forward contracts. Since bond contracts with maturities longer than 1 period contain an implicit commitment for actions in a future time period, we often find it helpful to make those implicit commitments explicit. For example, consider the position of a lender[12] in a bond contract with a maturity of 2 periods with a price today of $q_t^{(2)}$. The explicit commitment is lend an amount $q_t^{(2)}$ today in exchange for the guaranteed repayment of and amount 1 two periods in the future. Implicit in that contract, however, is a commitment from the lender to not terminate the loan early and demand repayment next period. That's equivalent to making you a 1-period loan today, then guaranteeing that next period I will extend you another 1-period loan in exchange for a guarantee to repay 1 at the end of that period.

[12] Remember a *lender* buys the bond so has a long position in the asset. A *borrower* sells the bond so has a short position in the asset.

But how big is that implicit loan for next period? Let's call that amount $g_t^{(1)}$ so we're asking how big is $g_t^{(1)}$?[13] I know that the repayment of 1 is the same, and also that the *implicit* contract has to be worth the same today as the *explicit* contract – we're not doing anything extra with this bond contract, we're just defining another way to do the accounting. This means that it must be the case that $q_t^{(2)} = q_t^{(1)} g_t^{(1)}$.

[13] The superscript on g is (1) which makes sences since I'll be making another 1-period loan next period. But notice that the time subscript on $g^{(1)}$ is t not $t+1$. Isn't that implicit loan made next period, so shouldn't the time subscript reflect that and therefore be $t+1$? Remember that time subscript is just our compact notation for when we observe information. This implicit loan is made at t therefore it must depend only on information available at that date and *not* information that we won't see until $t+1$.

The term we give to the implicit contract for the loan in the second period in this example is a *forward loan contract*, or simply a forward contract. And the price of this forward contract is implicit in the prices of the bonds of maturities $n = 1$ and $n = 2$:

$$g_t^{(1)} = q_t^{(2)} / q_t^{(1)}.$$

Forward rates. So far, all of our bonds have been purchases at some date t of claims to cash flows at futures dates $t + n$. A forward contract involves the agreement to purchase an asset in the future at a price agreed on now. Consider an agreement at date t to buy a 1-period bond at date $t + n - 1$. What should the price be? The agreement involves the purchase at $t + n - 1$ of a 1-period bond at the forward price of say $g_t^{(n-1)}$. The subscript t here is the date we agree on the terms of the agreement, not the date of execution of that agreement (which is $t + n - 1$). The one-period bond[14] delivers 1 at $t + n$.

[14] The notation gets a little bit confusing at this point since we now have two different maturities to keep track of. Since the forward contract settled $n - 1$ periods in the future, the *maturity* of the *forward* contract is $n - 1$. That is why the superscript in the price, $g_t^{(n-1)}$, is $n - 1$. The underlying bond that the forward contract delivers at $t + n - 1$ is a 1-period bond. That's implicit in the definition and isn't separately indicated in the notation. We could also think of forward contracts for bonds with longer maturities. In that case we would definitely want to change the notation to reflect both the maturity of the underlying bond and the maturity of the forward contract.

We can replicate these cash flows with long and short positions in bonds. What cashflows are we trying to replicate? I will need to pay $g_t^{(n-1)}$ at $t + n - 1$ and I will receive 1 at $t + n$. I know what the second cashflow costs today, since is just the price of an $t + n$ discount bond, $q_t^{(n-1)}$. But how much do I have to invest now to be sure that I have enough to pay $g_t^{(n-1)}$ at $t + n - 1$? I know if I buy an $n - 1$-period bond today, I will get back 1 at $t + n - 1$, which will cost me $q_t^{(n-1)}$. Therefore, if I buy $g_t^{(n-1)}$ units of that bond at a cost of $g_t^{(n-1)} q_t^{(n-1)}$, I get back just enough at $t + n - 1$ to pay the amount $g_t^{(n-1)}$. Since both of these strategies are guaranteed to deliver a risk-free payoff of 1 in $t + n$ and all other *net* cashflows are 0, the absence of arbitrage implies

$$\begin{aligned} q_t^{(n)} &= g_t^{(n-1)} q_t^{(n-1)} \\ \Rightarrow g_t^{(n-1)} &= q_t^{(n)} / q_t^{(n-1)}, \end{aligned}$$

which is a generalization to a longer maturity of n of what we had already knew for a 2-period bond.

The continuously-compounded interest rate on the 1-period bond underlying this forward contract – which we call the *forward rate* – is

$$\begin{aligned} f_t^{(n-1)} &= -\log g_t^{(n-1)} \\ &= \log q_t^{(n-1)} - \log q_t^{(n)}. \end{aligned}$$

There's nice recursive structure to this formula that allows us to write multi-period bond yields as averages of the forward rates of forward

contracts of different maturities (remember, the underlying bond maturity is always 1 period):

$$
\begin{aligned}
\log q_t^{(n)} &= \underbrace{\log q_t^{(n-1)}}_{=\log q_t^{(n-2)} - f_t^{(n-2)}} - f_t^{(n-1)} \\
&= \underbrace{\log q_t^{(n-2)}}_{=\log q_t^{(n-3)} - f_t^{(n-3)}} - f_t^{(n-2)} - f_t^{(n-1)} \\
&= \log q_t^{(n-3)} - f_t^{(n-3)} - f_t^{(n-2)} - f_t^{(n-1)} \\
&\vdots \\
&= -f_t^{(0)} - f_t^{(1)} - f_t^{(2)} - \ldots - f_t^{(n-1)} \\
\Rightarrow y_t^{(n)} &= \frac{f_t^{(0)} + f_t^{(1)} + f_t^{(2)} + \ldots + f_t^{(n-1)}}{n}.
\end{aligned}
$$

The last step follows from the definition of the yield, $y_t^{(n)} = (-1/n) \log q_t^{(n)}$.

Yields and forward rates are analogous to average and marginal cost in microeconomics: the yield measure the average cost per period of multi-period loan, whereas forward rates measure the incremental costs for each extra period of the multi-period loan.

Notice that $f_t^{(0)} = y_t^{(1)} = i_t$ which we call the *short rate*. That is, the rate of return on the shortest possible maturity, $n = 1$.

Evidently we can report bond prices $q_t^{(n)}$, yields $y_t^{(n)}$, or forward rates $f_t^{(n)}$. In each case we have one number for each maturity. They contain exactly the same information: once we know one of these sequences, we can compute the others. Forwards are mathematically cleaner, as we'll see, but yields are more common. Working with yields rather than forward rates is just a habit we've inherited from the past.

One last thing to remember: bond prices and returns have units. It's easy to lose sight of that when returns (or yields or forward rates) are reported as percentages. Typically the units are dollars (or other currency) – so-called *nominal* bonds – in which case bond prices are also measured in dollars: the price is the number of dollars you give up today to get a dollar in the future. Returns are also specific to those dollar units since they represent dollars later per dollar now.[15] We could use other units, but those are standard.

[15] Yields and forward rates have additional units given by our choice of compounding convention. We will always work with continuous compounding, but other types of compounding imply yields and forward rates measured in different units.

A quick look at some data

To get a feel for some of the properties of multiperiod discount bond prices, let's look at some actual data for the US. The closest we can get to a default-free asset in the real world is a US Treasury bond. People might speculate that the US government could possibly default on its debt obligations, but in reality the probability of that happening is so close to zero that we can safely treat US Treasury securities as default-free. Table 1 provides a summary of the types of bonds the US Treasury issues. We'll leave the last two aside for now

Table 1: Summary of U.S. Treasury Securities and Their Maturities

Security Type	**Maturities**
Treasury Bills	4, 8, 13, 26, and 52 weeks
Treasury Notes	2, 3, 5, 7, and 10 years
Treasury Bonds	20 and 30 years
Treasury Inflation-Protected Securities (TIPS)	5, 10, and 30 years
Floating Rate Notes (FRNs)	2 years

– we'll talk about inflation and TIPS at the end of this chapter, and FRNs involve options which we've talked about before – and focus on the three types.

Remember that the maturity listed in the table is only the maturity when the security is issues. All of these securities are traded in highly liquid secondary markets every day. In other words, in reality there is a virtual continuum of available maturities. For example, the US Treasury doesn't issue a 1-day T-Bill, but a 4-week maturity T-Bill issued on May 1st with a maturity date of May 31st will be traded in secondary markets as a 1-day maturity T-Bill on May 30th. And this is true for the whole spectrum of treasury issues.[16]

[16] You may see the term *constant maturity* bond used when data is combined in this way.

Treasury Bills are equivalent to the discount bonds we've been talking about, which is convenient. However, their maturities are relatively short – one year or less. We would like to understand the behavior of long-maturity discount bond prices – say 10 years or more – but longer maturity Treasury Notes and Treasury Bonds include periodic *coupon payments* as well as a face value. Longer-maturity discount bond prices, therefore, must be measured indirectly through these coupon bond prices.

Seeing how this works in practice turns out to be a nice application of arbitrage-free pricing, so it is worth mentioning briefly. Consider a 10-year coupon bond that had a face value of $1,000 that also paid

4% of the face value every year in two equal installments every 6 months, and that coupon bond is currently trading for a price of \$995. The cashflows for the full term of the bond would be \$20 every 6 months with the exception of the last cashflow at the end of year 10 which would be would be \$1,020. There's an easy way to replicate the cashflows of this bond using a portfolio of pure-discount bonds with face values of \$1, and with maturities (in months) of $n = 6, 12, 18, \ldots, 120$: I would buy 20 of each of these bonds, then I would buy an addition 1,000 of the $n = 120$ bond. What would the cashflows from owning this portfolio look like? Notice that every 6 months some of the discount bonds in this portfolio mature, pay \$1, then disappear, while the others live on. The cashflows are therefore \$20 every 6 months with the exception of the last cashflow of year 10 which would be would be \$1,020, which is exactly the same as the coupon bonds cashflows. In the absence of arbitrage, the portfolio of discount bonds and the coupon bond must have exactly the same price:

$$\$995 = 20q_t^{(6)} + 20q_t^{(12)} + 20q_t^{(18)} + \ldots + 1,020q_t^{(120)}.$$

This argument generalizes to any default-free bond with a face value of F, guaranteed coupon payments of $\{C_1, C_2, \ldots, C_n\}$, and a date-$t$ price of B_t:

$$B_t = \sum_{i=1}^{n} C_i q_t^{(i)}.$$

Data on at least n of these bonds, *i.e.* the coupon bond price along with the sequence of dividends, must satisfy the n linear equations in n unknowns, $q_t^{(i)}$), defined above. The solution to that linear system of equations are then the arbitrage-free prices of default-free, pure-discount bonds of all maturities up to n. There are some subtleties to this, as there is with any measurement exercise, but generally speaking, that's what we do in practice.

Table 2 contains actual discount bond yields for for maturities ranging from 1 to 40 months for each quarter of 1963. Yields on longer-maturity bonds are typically higher than shorter-maturity bonds, and there is substantial variation in those yields across the 4 quarters of 1963. It's easier to see these patterns by plotting yields against their maturities – remember we call this plot the *yield curve* – which we do in Figure 1.

The shape of the yield curve in 1963 was upward sloping in each quarter, exhibited more variation across time at the short end of

Maturity		**Yields (%CCAR)**		
(months)	1963Q1	1963Q2	1963Q3	1963Q4
1	2.7817	2.6784	2.8254	3.1822
2	2.8162	2.7443	2.8679	3.2190
3	2.8588	2.8179	2.9253	3.2720
4	2.9065	2.8936	2.9903	3.3309
5	2.9573	2.9679	3.0581	3.3900
6	3.0095	3.0390	3.1257	3.4465
7	3.0620	3.1062	3.1911	3.4990
8	3.1139	3.1688	3.2533	3.5470
9	3.1646	3.2269	3.3116	3.5905
10	3.2137	3.2806	3.3659	3.6297
11	3.2608	3.3301	3.4161	3.6649
12	3.3059	3.3758	3.4624	3.6967
13	3.3488	3.4179	3.5050	3.7253
14	3.3896	3.4566	3.5441	3.7511
15	3.4282	3.4924	3.5800	3.7744
16	3.4647	3.5255	3.6130	3.7956
17	3.4991	3.5561	3.6434	3.8148
18	3.5316	3.5844	3.6713	3.8324
19	3.5622	3.6106	3.6970	3.8485
20	3.5911	3.6350	3.7208	3.8633
21	3.6183	3.6577	3.7427	3.8768
22	3.6439	3.6789	3.7630	3.8893
23	3.6681	3.6986	3.7819	3.9009
24	3.6909	3.7170	3.7994	3.9116
25	3.7124	3.7343	3.8157	3.9216
26	3.7327	3.7505	3.8310	3.9309
27	3.7520	3.7656	3.8452	3.9395
28	3.7701	3.7799	3.8585	3.9476
29	3.7873	3.7934	3.8710	3.9552
30	3.8036	3.8060	3.8827	3.9623
31	3.8190	3.8180	3.8937	3.9690
32	3.8337	3.8293	3.9041	3.9753
33	3.8476	3.8399	3.9139	3.9812
34	3.8608	3.8500	3.9232	3.9868
35	3.8734	3.8596	3.9319	3.9921
36	3.8853	3.8687	3.9402	3.9971
37	3.8967	3.8774	3.9481	4.0019
38	3.9076	3.8856	3.9555	4.0064
39	3.9180	3.8935	3.9626	4.0107
40	3.9279	3.9009	3.9694	4.0148

Table 2: Here's some data constructed and published by the Board of Governors of the US Federal Reserve System. (Source: Federal Reserve Board.) These correspond to maturities that range from 3-months to 10 years in 3-month intervals (*i.e.*, n=1,2,3,...,40, measured in quarters of a year). And to get a feel for how prices change over time, we measure prices every quarter from 1962 to 2016. Note that the quarterly frequency and the quarterly maturities are arbitrary choices. If we were interested in higher-frequency price movements, we could measure prices daily, for example, and if we are interested in other maturities, we would solve a slightly different set of linear equations. Here's what yields looked in 1963.

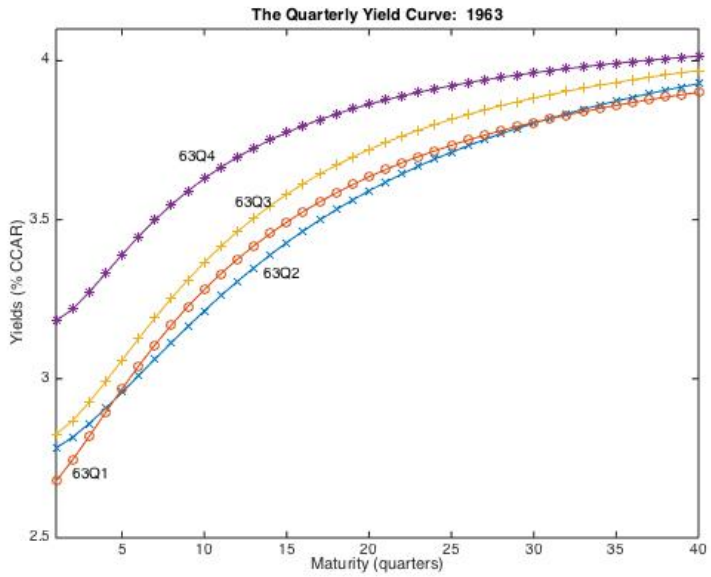

Figure 1: For each of four quarterly time periods in Table 2, the yields on default-free pure-discount bonds for maturities of 1 quarter up to 40 quarters (10 years) is plotted against its maturity. Percentage yields are continuously compounded at annual rates (CCAR).

maturities than at the long end, and the shape of the curve changed a little bit over time (you will notice that the yield curves for different quarters intersect in a few places, reflecting that the curve doesn't exactly move up and down in parallel).

Was 1963 typical, or on average, does the yield curve have substantially different properties? A 3D plot of yield curves for each quarter from 1962 to 2016 in Figure 2 displays the full sample. This is pretty to look at, but I personally find 3D graphs challenging to analyze. There are a few broad patterns that we can see in this graph. It looks like yield curves we low in the 1960s then tended to drift up until the late 1970s, then they began to slowly come down to their current low levels. They mostly slope up (that's looking into the 3rd dimension), but there also seem to be a few periods where the shape is reversed and yield curves actually slope down.

Let's summarize this information the traditional way. Figure 3 plots the averages across all of the quarters from 1962 to 2016 for each maturity, along with the standard deviation. Indeed the average yield curve slopes up ranging from just over 5% for a 3-month maturity to just under 6.5% for a 10-year maturity (note that the units on the vertical axis are "% CCAR" which is a common abbreviation for "percentage continuously compounded at annual rates"). And there is substantial variation in yields across time. The standard deviation

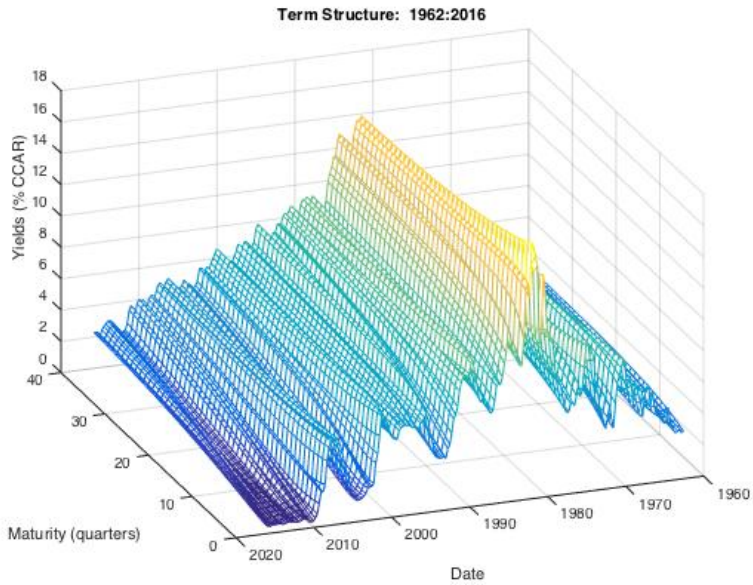

Figure 2: For each of quarter from 1962 to 2016, the yields on default-free pure-discount bonds for maturities of 1 quarter up to 40 quarters (10 years) is plotted against its maturity. Percentage yields are continuously compounded at annual rates (CCAR).

for a 3-month yield is about 3.25%, falling to just over 2.5% for yields on bonds with a 10-year maturity.

To get a sense of how the shape of the yield curve changes over time, Figure 4 plot the time series for the 3-month and the 10-year bonds on the same graph. When the 10-year is above the 3-month, as is typically the case as we know from looking at the sample average, then the yield curve is sloping up. But there are a few periods where that pattern is reversed: the 3-month yield is bigger than the 10-year yield implying the yield curve has a negative slope (*e.g.*, 1974, 1982, 1983, 1990, 2001 and 2007). Another thing that is evident from these time series plots is that yields are very persistent: when they are high then tend to stay high, and when they're low they tend to stay low. Their not perfectly persistent, of course, but both the short and long bond yields exhibit a strong positive auto-correlation. (We'll leave the statistical estimation of that property for later.)

Recall that there were 3 equivalent ways to measure bond prices. In addition to yields, we could also look at the prices themselves, which we plot in Figure 5. It's harder to see patterns in this price curve than it was the yield curve, which is why we tend to favor the per-period units of yields rather than the more complicated multi-period units of prices.

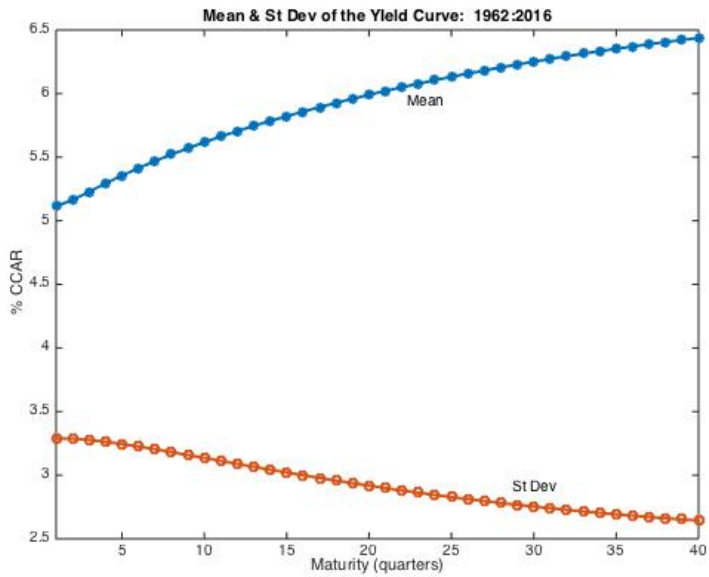

Figure 3: Average yields, $\bar{y}^{(n)} = \Sigma_{t=1}^{T} y_t^{(n)}$, and standard deviation, $[\Sigma_{t=1}^{T}(y_t^{(n)} - \bar{y}^{(n)})/T]^{1/2}$, for quarterly data from 1962 to 2016.

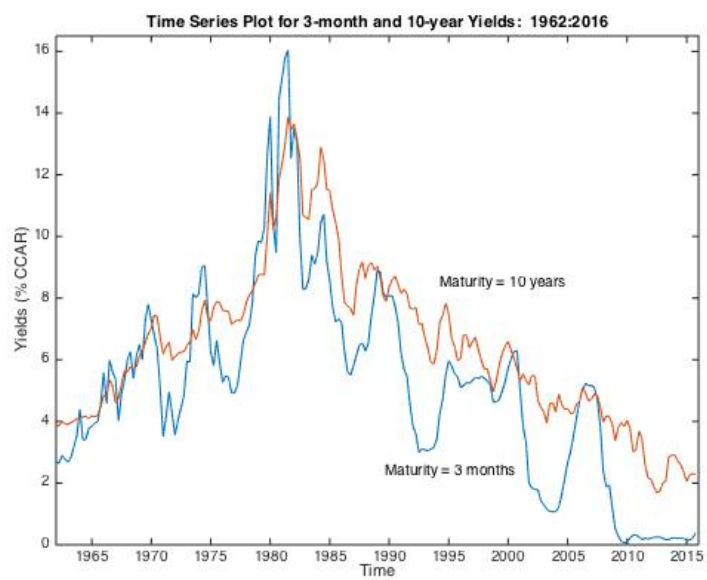

Figure 4: Time series plot of yield each quarter from 1962 to 2016 is plotted against, for a 3-month and a 10-year bond. Percentage yields are continuously compounded at annual rates (CCAR).

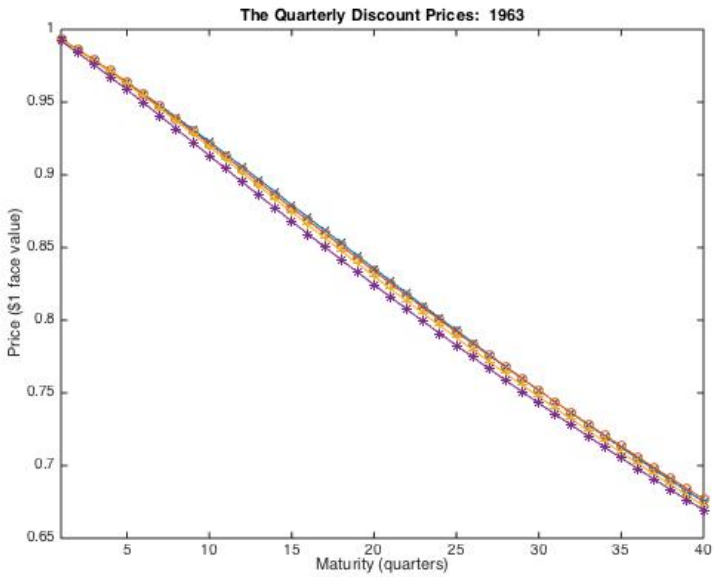

Figure 5: Price in dollars for a default-free pure-discount bond that pays one dollar at maturity. There is one plot for each quarter of 1963.

The third option for displaying these data is the *forward curve*, *i.e.*, the plot of forward rates against maturity. As you can see in Figure 6, the patterns displayed in the yield curves for the 4 quarters of 1963 are even more apparent in the forward curves. Recall the interpretation of forward rates as "marginal costs" rather than the "average cost" interpretation of yields. One interesting pattern is the apparent convergence in forward rates: the forward curves have a steep slope at the short end, but then quickly flatten out around 5 years, and seem to converge on a constant by 10 years.

How does 1963 compare to averages for the entire sample period? Figure 7 plots of the average forward curve along with its standard deviation. The average forward curve also slopes up and exhibits substantial variability at all maturities. Compared to the average yield curve, the average forward curve is indeed steeper at short maturities and flatter at long maturities, and has variability that decreases faster as maturity increases.

We next develop some bond-pricing theory that will help us interpret these apparent patterns in the data. We'll then return to the data to both calibrate the constant parameters of our models, and to evaluate their overall quantitative properties.

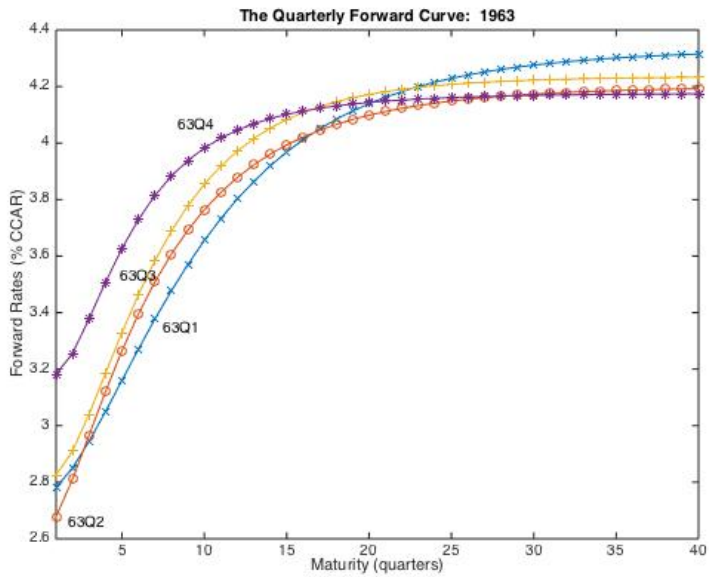

Figure 6: For each of quarter from 1962 to 2016, the forward rate implied by default-free pure-discount bonds for maturities of 1 quarter up to 40 quarters (10 years) is plotted against its maturity. Percentage rates are continuously compounded at annual rates (CCAR).

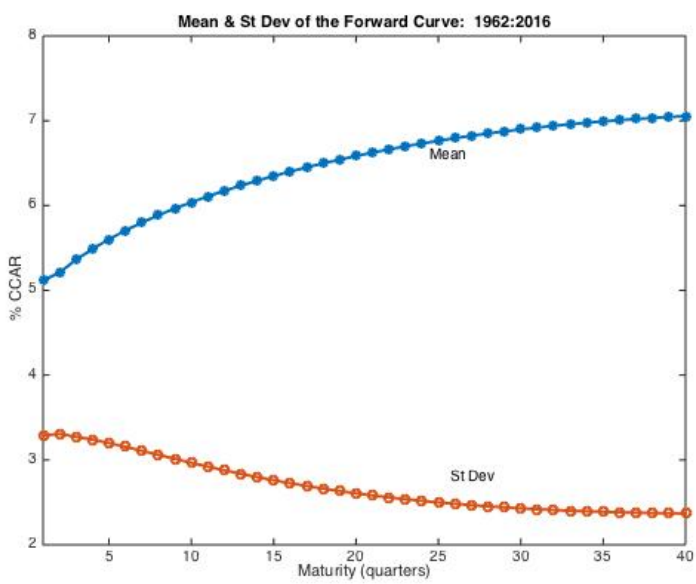

Figure 7: Average forward rates, $\bar{f}^{(n)} = \Sigma_{t=1}^{T} f_t^{(n)}$, and standard deviation, $[\Sigma_{t=1}^{T}(f_t^{(n)} - \bar{f}^{(n)})/T]^{1/2}$, for quarterly data from 1962 to 2016.

Pricing kernels and bond prices

The simplicity and elegance of bond pricing comes from the lack of an uncertain dividend. Since the cash flows are known – that's why we refer to bonds and related securities as *fixed income* – prices depend only on the pricing kernel.[17] To make the point absolutely clear: the pricing kernel alone determines bond prices. Period.

[17] Defaultable bonds are also often called *fixed-income* assets. That terminology is slightly different since in that case, it is the *promise* to pay that is fixed, not necessarily the ultimate payment itself.

We value assets as we did before: using the no-arbitrage theorem. In a dynamic setting, this is equation (1). The difference with our earlier work is that m is now a stochastic process rather than a simple timeless random variable.

[Reminder: recall that the pricing kernel has inverse units to a return. If r_{t+1} is dollars tomorrow for dollars today, then m_{t+1} has units of 1/dollars tomorrow for 1/dollars today. That way the units cancel and we're left with the unit-free "1" on the right-hand side of the fundamental equation of asset pricing.]

What are the *bond returns* we plug into this pricing relation? The relevant returns are sometimes called *holding period returns* for a holding period of one. If we buy an $(n+1)$-period bond at t, hold it for one period then sell it, we would have paid $q_t^{(n+1)}$ and gotten back $q_{t+1}^{(n)}$. Notice that the maturity is shorter by one period since we are now one-period closer to maturity when we sell than when we bought the bond. Our (gross) return is

$$r_{t+1}^{(n+1)} = q_{t+1}^{(n)} / q_t^{(n+1)}. \tag{2}$$

Same object for bonds as the returns we've computed on other assets.

Now put this to work. If we substitute the return (2) into the pricing relation (1) we have

$$q_t^{(n+1)} = E_t\left[m_{t+1} q_{t+1}^{(n)}\right]. \tag{3}$$

So if we know the price of an n-period bond, we can compute the price of an $(n+1)$-period bond. We do this recursively starting with $q_t^{(0)} = 1$ (which just says that \$1 paid right now is worth \$1). The first one is what we've been doing all along:

$$q_t^{(1)} = E_t\left[m_{t+1} q_{t+1}^{(0)}\right] = E_t\left[m_{t+1}\right].$$

Long-maturity bonds follow from the same logic repeated: given the price of an n-period bond, use (3) to find the price of an $n+1$-period bond.

We've done this recursively, but you can imagine doing it "all at once." The price of a two-period bond, for example, can be written

$$\begin{aligned} q_t^{(2)} &= E_t\left[m_{t+1} q_{t+1}^{(1)}\right] \\ &= E_t\left[m_{t+1} \underbrace{E_{t+1}[m_{t+2}]}_{=q_{t+1}^{(1)}}\right] \\ &= E_t[m_{t+1} m_{t+2}]. \end{aligned}$$

The last line follows from the law of iterated expectations. Repeating the same logic, we find

$$q_t^{(n)} = E_t\left[m_{t+1} m_{t+2} \cdots m_{t+n}\right].$$

In other words, $m_{t+1} m_{t+2} \cdots m_{t+n}$ is the pricing kernel we would use to discount from $t+n$ back to t. [18]

[18] If this seems overly mysterious, but you understand the logic of the recursive relationship in equation (3), then just leave it alone for now. Later you might want to review the *law of iterated expectations* we introduced with stochastic processes.

Why do we focus on the dynamics of m?

It might seem odd to look at bond prices at a single point in time, and claim that those prices are driven by the dynamics of the pricing kernel, *i.e.*, not the pricing kernel at that point in time, but rather its evolution across many time periods.

An iid pricing kernel. To make the point that pricing-kernel dynamics are what drives bond prices, consider the case of a pricing kernel whose random properties are *independently and identically distributed* over time, *i.e.*, a pricing kernel with no dynamics. The one-period bond price is, as we've seen,

$$q_t^{(1)} = E_t\left[m_{t+1}\right].$$

In general, this price changes over time because information arrives and changes the properties of the distributions of future outcomes. Investor's internalize this by using a conditional expected value – conditional on current information in an environment in which we understand that information changes over time. Since we're considering the case where m_{t+1} is iid, there is no information available that changes this expected value over time – that's the meaning of *independently* in *iid*, the distribution doesn't depend on anything that came before and doesn't affect anything that will come after. In this case, the bond price is just the same constant number, $q^{(1)}$, in all date-t states and for all time periods,

$$q_t^{(1)} = E_t\left[m_{t+1}\right] = E\left[m_{t+1}\right] = q^{(1)}.$$

Since this price never changes, $q_{t+1}^{(1)} = q^{(1)}$, that same constant number, so for $n = 2$ we have

$$q_t^{(2)} = E_t\left[m_{t+1} q_{t+1}^{(1)}\right] = q^{(1)} E\left[m_{t+1}\right] = q^{(1)} q^{(1)}.$$

Using the same logic over and over again gives us

$$q_t^{(n)} \quad = \quad (q^{(1)})^n,$$

which is also constant for all date-t states and for all time periods.

This implies that the yield curve is perfectly flat and never changes, *i.e.*, yields on all maturities are exactly the same, equal to the constant short rate for all time:

$$y_t^{(n)} = (-1/n)\log q_t^{(n)} = (-1/n)\log[(q^{(1)})^n] = -\log q^{(1)} = i.$$

Likewise for the forward curve:

$$\begin{aligned} f_t^{(n)} &= \log q_t^{(n)} - \log q_t^{(n+1)} \\ &= \log[(q^{(1)})^n] - \log[(q^{(1)})^{n+1}] \\ &= n\log q^{(1)} - (n+1)\log q^{(1)} \\ &= -\log q^{(1)} \quad = \quad i. \end{aligned}$$

Any way you look at it, there's no interest rate risk if m is iid: both the yield curve and forward curve are equal and perfectly flat, and they do not vary across time. Recall that this is nothing like what we see in the data. In other words, if we expect to have a theoretical model that is helpful for understanding those data, then we need to focus on the dynamics of m.

But what kind of dynamics? The fact that the average yield curve slopes up is actually telling us something about the autocorrelation of the pricing kernel. And the fact that long maturity forward rates have almost as large a variance as they do at short maturities, is telling us something the persistence of shocks to the pricing kernel.

Consider the risks associated with buying a 2-period bond. Even though there's no risk associated with the bond's payoff 2-periods into the future, it's payoff next period is still random. Recall that after one period elapses, a 2-period bond is itself just a 1-period bond, but with a price that won't be known until next period. That price is the risky cashflow associated with the 2-period bond:

$$q_t^{(2)} \quad = \quad E_t[m_{t+1}q_{t+1}^{(1)}].$$

By the law of iterated expectations this price can be written in terms of 1-period pricing kernels,

$$\begin{aligned} q_t^{(2)} &= E_t[m_{t+1}\overbrace{E_{t+1}m_{t+2}}^{=q_{t+1}^{(1)}}] \\ &= E_t[m_{t+1}m_{t+2}] \\ &= (E_t m_{t+1})(E_t m_{t+2}) + \mathrm{Cov}_t(m_{t+1}, m_{t+2}). \end{aligned}$$

In other words, the 2-period bond represents a risky investment whenever $\text{Cov}_t(m_{t+1}, m_{t+2}) < 0$. Intuitively, when m_{t+1} is relatively high, the negative correlation says that m_{t+2} will be relatively low, which in turn implies that the 1-period bond price next period will also be relatively low. In other words, the cash flow from the 2-period bond will be low when the pricing kernel is high (and *vice versa*). This might be easier to see in a simple example.

Example: an AR(1) log pricing kernel. Without any deeper motivation, let's make an assumption about the dynamics of the pricing kernel and work out some simple properties of bond prices and yields. In particular, assume that the log of the pricing kernel follows an AR(1) model:

$$\log m_{t+1} \quad = \quad \mu(1-\varphi) + \varphi \log m_t + \sigma w_{t+1},$$

where μ, φ, and σ are constant parameters, and $w_t \sim iid\mathcal{N}(0,1)$. As we've seen, the dynamics are driven by the parameter φ. Recall that for this AR(1) model, the distribution of $\log m_{t+n}$ conditional on $\log m_t$ is,

$$\log m_{t+n} \mid \log m_t \quad \sim \mathcal{N}\Big((1-\varphi^n)\mu + \varphi^n \log m_t \,,\, \sigma(1-\varphi^{2n})/(1-\varphi^2) \Big).$$

Therefore, the closer $|\varphi|$ is to 1, the bigger is the effect of the current value of $\log m_t$ on the mean and variance of the distributions for future values of $\log m_{t+n}$. And when $|\varphi|$ is very close to 1, this will be true even when n is quite large.

To see that both the sign and the magnitude of φ determine the properties of the mean yield curve, we can find the mean interest rate and the means of longer-maturity bond yields as functions of φ. Let's start with the interest rate:

$$\begin{aligned}
i_t &= -\log E_t\left[e^{\log m_{t+1}}\right] \\
&= -\log E_t\left[e^{\mu(1-\varphi)+\varphi \log m_t + \sigma w_{t+1}}\right] \\
&= -\mu(1-\varphi) - \sigma^2/2 - \varphi \log m_t \\
\Rightarrow Ei_t &= -\mu(1-\varphi) - \sigma^2/2 - \varphi \underbrace{E(\log m_t)}_{=\mu} \quad = \quad -\mu - \sigma^2/2.
\end{aligned}$$

What about the mean of longer-maturity bonds? Let's keep things

simple and focus on the $n = 2$ case:

$$\begin{aligned}
y_t^{(2)} &= -(1/2)\log E_t\left[e^{\log m_{t+1}+\log m_{t+2}}\right] \\
&= -(1/2)\log E_t\left[e^{\mu(1-\varphi)+\varphi\log m_t+\sigma w_{t+1}+\mu(1-\varphi)+\varphi\log m_{t+1}+\sigma w_{t+2}}\right] \\
&= -(1/2)\log E_t\left[e^{\mu(1-\varphi)(2+\varphi)+(\varphi+\varphi^2)\log m_t+(1+\varphi)\sigma w_{t+1}+\sigma w_{t+2}}\right] \\
&= -(1/2)\mu(1-\varphi)(2+\varphi) - (1/2)(1+\varphi)^2+1]\sigma^2/2 - (1/2)(\varphi+\varphi^2)\log m_t \\
\Rightarrow Ey_t^{(2)} &= -\mu - (1/2)[(1+\varphi)^2+1]\sigma^2/2.
\end{aligned}$$

Upward-sloping average yield curve. If we observe $Ey_t^{(2)} > Ei_t$ (as we do for real market data), then it must be the case that

$$\begin{aligned}
0 &< \underbrace{-\mu - (1/2)[(1+\varphi)^2+1]\sigma^2/2}_{=Ey_t^{(2)}} - (\underbrace{-\mu - \sigma^2/2}_{=Ei_t}) \\
\Rightarrow 1 &> [(1+\varphi)^2+1]/2 \\
\Rightarrow \varphi &< 0.
\end{aligned}$$

In other words, the log pricing kernel must be negatively autocorrelated. We could work out the means of yields when $n > 2$ and show that $\varphi < 0$ is sufficient for the entire mean yield curve to slope up. But we'll save that for the more interesting pricing kernel models we develop below.

Variability of long-maturity forward rates. If we had worked out long-maturity bond prices for this simple AR(1) pricing kernel model, we would've found that they imply forward rates for maturity n that have the form

$$f_t^{(n-1)} = (\text{constant function of } n) + \varphi^n \log m_t,$$

which implies

$$\mathrm{Var}(f_t^{(n-1)}) = \varphi^{2n}\,\mathrm{Var}(\log m_t).$$

In other words, the variance of the forward rate is converging to zero exponentially fast as maturity grows, with φ^2 determining the rate of convergence. For example, if we think of a period as a quarter of a year and if $\varphi = -0.5$, then the forward rate implied by a 10-year maturity would have a variance that is equal to the variance of the log pricing kernel multiplied by a number that is zero to 24 decimal places! This means that either the log pricing kernel is unbelievable variable – it's not, of course – or the value of φ^2 must be much much closer to 1. On the other hand, if $\varphi = -0.95$ suggesting much higher persistence, then the variance of the pricing kernel is scaled by 0.0165, which seems like a much more reasonable number.

This intuition – empirical properties of the yield curve suggest a pricing kernel with negative autocorrelation and a lot of persistence – will extend beyond this simple AR(1) example. But the analytics that would establish those facts get a much more involved. We develop those analytics more systematically below. We'll start simple, as usual, and add complexity only when necessary.

Markov models

Let's start with a Markov model. A state variable x_t evolves according to some well-behaved stochastic process. The pricing kernel between t and $t+1$ is a function of both states, $m(x_t, x_{t+1})$.

What about bond prices? The one-period bond price is evidently a function of the current state:

$$q^{(1)}(x_t) \quad = \quad E_t\big[m(x_t, x_{t+1})\big].$$

Why doesn't it depend on x_{t+1}? Because we've taken the expectation over x_{t+1} (mathematically, we've integrated over the x_{t+1} dimension, which is just a mathy way of saying we've taken a probability-weighted average across all possible values of x_{t+1}.). Similar logic applies to bonds of longer maturity:

$$q^{(n+1)}(x_t) \quad = \quad E_t\big[m(x_t, x_{t+1})q^{(n)}(x_{t+1})\big].$$

We're looking, therefore, for the functions $q^{(n)}(x_t)$. Once we have them, we can compute yields and forward rates, which are also functions of the state.

The Vasicek model

The Vasicek model[19] has a convenient loglinear structure, and it's a good illustration of how bond-pricing models work. We describe the model and go on to connect its parameters to what we know about the behavior of interest rates. The model delivers bond prices whose logs are linear in the state variable x_t, which means yields and forward rates are linear in the state as well.

Model. The Vasicek model has two components. One is an autoregressive state variable,

$$x_{t+1} \quad = \quad \delta_x(1-\varphi) + \varphi x_t + \sigma_x w_{t+1},$$

[19] It is now common practice to anglicize the spelling of the Czech name Vašíček. The original research was published way back in 1977: Vasicek, O., "An equilibrium characterization of the term structure," *Journal of Financial Economics* 5, 177–188. The model in that paper is written in continuous time, but it translates exactly to our discrete-time framework. Also, in case you think this is just a mathematical exercise, Vasicek built a financial consulting company, KMV, based largely on applications of this model – and extensions to defaultable bonds – which he and his partners sold to Moody's in 1989 for $210 million (that's about $450 million in 2020 dollars).

where $\{w_t\}$ is the usual collection of independent standard normals. The other is a loglinear pricing kernel,

$$\log m(x_t, x_{t+1}) \quad = \quad a + bx_t + cx_{t+1},$$

where a, b, and c, are constant parameters. to get a clearer picture of what's implied by this stochastic process for the pricing kernel, we substitute for x_{t+1} to get

$$\begin{aligned} \log m(x_t, w_{t+1}) &= a + bx_t + c\overbrace{[\delta_x(1-\varphi) + \varphi x_t + \sigma_x w_{t+1}]}^{=x_{t+1}} \\ &= a + c\delta_x(1-\varphi) + (b + c\varphi)x_t + c\sigma_x w_{t+1}. \end{aligned}$$

This is fine, of course, but it has too many parameters. We won't ever be able to tell the difference between b and c or the difference between c and σ_x.[20] Note that since at this point, x_t is an *abstract* state variable – meaning we haven't attached any particular economic interpretation to x_t – we're free to choose whatever units we like for it. That is, until we attach more meaning to x_t, we can't say if it more likely to take on positive or negative values (this depends on the sign and magnitude of δ_x), or when it's changes are big or small (this depends on the magnitude of σ_x). Redefine the state variable by multiplying it by a constant (recall that whenever we do this, we'ver implictly changing the units of that variable):

[20] In statistics we would say that these parameters are not separately *identified*.

$$z_t \equiv -(b + c\varphi)x_t.$$

We haven't gained or lost or changed any information, we've just changed the units of measurement. The AR(1) model for z_t follows this definition:

$$\begin{aligned} z_t &= -(b + c\varphi)x_t \\ &= -(b + c\varphi)[\delta_x(1-\varphi) + \varphi x_{t-1} + \sigma_x w_t] \\ &= -(b + c\varphi)\delta_x(1-\varphi) \underbrace{-\varphi(b + c\varphi)x_{t-1}}_{=\varphi z_{t-1}} - (b + c\varphi)\sigma_x w_t \\ &= \delta(1-\varphi) + \varphi z_{t-1} + \sigma w_t, \end{aligned}$$

where we have implicitly defined new mean and variance parameters for z_t as $\sigma = -(b + c\varphi)\sigma_x$ and $\delta = -(b + c\varphi)\delta_x$ to reflect the change in units.

Now lets go back and substitute this into the pricing kernel so we can re-write it as a function of the state variable z_t and future shocks to that state variable w_{t+1}:

$$\begin{aligned} \log m(z_t, w_{t+1}) &= a + c\delta_x(1-\varphi) + (b + c\varphi)\overbrace{[-z_t/(b + c\varphi)]}^{=x_t} + c\sigma_x w_{t+1} \\ &= -\lambda^2/2 - z_t + \lambda w_{t+1}, \end{aligned}$$

where the last line introduces yet another convenient normalization: we've chosen an arbitrary normalization $b + c\varphi = -1$ and $a + c\delta_x(1 - \varphi) = -\lambda^2/2$, where $\lambda = c\sigma$. This last step looks a little suspicious, but it's really just choosing convenient notation.

The parameter λ gets its own name. We call it the *price of risk* since it measures the percentage change in the pricing kernel for an incremental change in risk: $\lambda = \partial \log m_{t+1}/\partial w_{t+1}$. (Note that since z_t is known at date-t it's no longer risky, therefore it's the *shocks* that arrive in the future, w_{t+1}, that embody risk.)

The probability distribution of $\log m_{t+1}$ conditional on a particular value of z_t, *i.e.*, the *conditional distribution*, is normal with a conditional mean $-\lambda^2/2 - z_t$ and conditional variance λ^2,

$$\log m_{t+1} \mid z_t \sim \mathcal{N}\big(-\lambda^2/2 - z_t\,,\ \lambda^2\big),$$

which implies that the cumulant generating function for the pricing kernel conditional on z_t has a particularly simply form:

$$\begin{aligned} k_{m,t}(s) &= \log E\big[e^{s \log m_{t+1}} \mid z_t\big] \\ &= s(-\lambda^2/2 - z_t) + (s^2/2)\lambda^2, \end{aligned}$$

where we've added a subscript m to denote the variable associated with this cgf, and the time script t do denote we are using a conditional distribution that holds fixed the value z_t.

Why do we make this change in scale/notation? It makes the short rate exactly equal to z_t:

$$i_t = -\log E_t[e^{\log m_{t+1}}] = -k_{m,t}(1) = z_t.$$

In other words, we've scaled the *abstract* state variable in such a way that it is no longer abstract. It now has a tangible *economic* interpretation as the short rate. The Vasicek model now can be summarized with 2 components: an AR(1) stochastic process for the short rate, and a price-of-risk parameter λ:

$$\begin{aligned} i_t &= \delta(1 - \varphi) + \varphi i_{t-1} + \sigma w_t \\ \log m_{t+1} &= -\lambda^2/2 - i_t + \lambda w_{t+1}. \end{aligned}$$

We will see the convenience of this representation when we use real data to inform our choices for the four parameters of this model, δ, φ, σ, and λ.

Note the intuitive nature of this model. Discounting future cashflows with the interest rate is how we first learn about the *time-value of*

money. Our pricing kernel then adds a model of risk valuation to this basic discounting:

$$m_{t+1} \quad = \quad e^{-i_t} \quad \underbrace{e^{-\lambda^2/2+\lambda w_{t+1}}}_{\text{risk adjustment}} \quad .$$

One more thing. The dynamics for the state variable i_t are an AR(1). But the dynamics for the pricing kernel are slightly more complicated. If we substitute the infinite-order moving-average representation[21] for i_t, we see that $\log m_{t+1}$ is

$$\log m_{t+1} \quad = \quad -\lambda^2/2 - \delta + \lambda w_{t+1} - \sigma \sum_{j=0}^{\infty} \varphi^j w_{t-j}. \tag{4}$$

[21] Recall that since i_t is an AR(1), it's moving-average representation is $i_t = \delta + \sigma \sum_{j=0}^{\infty} \varphi^j w_{t-j}$.

Since the coefficients decline at a geometric rate after the first one, we know this is an ARMA(1,1). Alternatively, we could substitute for the pricing kernel in the AR(1) for the interest rate:

$$\underbrace{(-\log m_{t+1} - \lambda^2/2 + \lambda w_{t+1})}_{=i_t} \quad = \quad \delta(1-\varphi) + \varphi(\underbrace{-\log m_t - \lambda^2/2 + \lambda w_t}_{=i_{t-1}}) + \sigma w_t$$

which implies

$$\log m_{t+1} \quad = \quad -(\delta + \lambda^2/2)(1-\varphi) + \varphi \log m_t + \lambda w_{t+1} - (\sigma + \varphi\lambda) w_t,$$

which is the Markov form for an ARMA(1,1). The autocorrelation of the log pricing kernel is now more complicated than the simple AR(1) example we did earlier. Recall the lesson of that simple model was that an upward-sloping average yield curve required a negatively autocorrelated log pricing kernel. And recall that for this ARMA(1,1) model, the first-order autocorrelation will be

$$\text{Corr}(\log m_{t+1}, \log m_t) \quad = \quad \frac{-(\lambda\sigma + \lambda^2\varphi)[1 + \varphi/(1-\varphi^2)]}{\lambda^2 + \sigma^2/(1-\varphi^2)}.$$

and for higher-order lags, $n > 1$,

$$\text{Corr}(\log m_{t+n}, \log m_t) \quad = \quad \varphi^n \text{Corr}(\log m_{t+1}, \log m_t).$$

In other words, the sign of the autocorrelations of the log pricing kernel will be determined jointly by the signs (and magnitudes) of λ and φ. We will see below that the interest rate is highly positively autocorrelated, which implies φ is positive and close to 1. Therefore, $\text{Corr}(\log m_{t+1}, \log m_t) < 0$ requires $\lambda > 0$ – risk has a positive price.

Recursions. We build up bond prices one at a time from the pricing relation

$$q_t^{(n+1)} \quad = \quad E_t\left[m_{t+1} q_{t+1}^{(n)}\right],$$

starting with $q_t^{(0)} = 1$ (a dollar now is worth a dollar). We already know about the short rate,

$$q_t^{(1)} \quad = \quad E_t\big[m_{t+1}\big] \quad = \quad e^{-i_t}.$$

For bonds of longer maturity, we guess prices have a loglinear[22]

$$\log q_t^{(n)} \quad = \quad A_n + B_n i_t$$

for some coefficients $\{A_n, B_n\}$ to be determined. The subscripts on A and B reflect the fact that these coefficients will be different for each maturity. You can think of this as a just a lucky guess, or you could work out the algebra by hand for $n = 2$, $n = 3$, etc, until you realized that the log of prices will always be linear in i_t, but with intercept and slope that change with n. The guess-and-verify method is much less work, and will apply to any value of n.

[22] You may hear this called "exponential-affine" rather than "loglinear". The exponential vs. log part is obvious: $y = f(x)$ where $\log f(x) = a + bx$ or $f(x) = e^{a+bx}$. The term "affine" rather than "linear" is a bit more obscure. The distinction is relevant only in a context in which all "linear" functions go through the origin, *i.e.*, $a = 0$. If that's the meaning given to "linear" then we need another name for the $a \neq 0$ case, *i.e.*, "affine".

Think about this for some particular maturity $n+1$. The pricing relation tells us we need to find the conditional mean of the product $m_{t+1}q_{t+1}^{(n)}$. The log of this conditional mean is simply the conditional cumulant generating function of the random variable $\log\left[m_{t+1}q_{t+1}^{(n)}\right]$ evaluated at $s = 1$. To see this, Take logs, substitute, and simplify:

$$\begin{aligned}
\log m_{t+1} + \log q_{t+1}^{(n)} \quad &= \quad \overbrace{-\lambda^2/2 - i_t + \lambda w_{t+1}}^{=\log m_{t+1}} + \overbrace{A_n + B_n \underbrace{\big[\delta(1-\varphi) + \varphi i_t + \sigma w_{t+1}\big]}_{=i_{t+1}}}^{=\log q_{t+1}^{(n)}} \\
&= \quad \big[-\lambda^2/2 + A_n + B_n\delta(1-\varphi)\big] + (\varphi B_n - 1)i_t + (\lambda + B_n\sigma)w_{t+1}.
\end{aligned}$$

Therefore, the conditional-on-i_t distribution is given by

$$\big(\log m_{t+1} + \log q_{t+1}^{(n)}\big) \mid i_t \sim \mathcal{N}\Big(\big[-\lambda^2/2 + A_n + B_n\delta(1-\varphi)\big] + (\varphi B_n - 1)i_t\,,\ (\lambda + B_n\sigma)^2\Big),$$

so the conditional cumulant generating function is

$$\begin{aligned}
k_{n+1,t}(s) \quad &= \quad \log E\big[e^{s\left(\log m_{t+1} + \log q_{t+1}^{(n)}\right)} \mid i_t\big] \\
&= \quad s\big[-\lambda^2/2 + A_n + B_n\delta(1-\varphi) + (\varphi B_n - 1)i_t\big] + (s^2/2)\big[(\lambda + B_n\sigma)^2\big],
\end{aligned}$$

where we have put a subscript $n+1$ on the cgf to denote that this is related to an $n+1$-maturity bond, and also the customary time subscript t to denote the use of the conditional distribution holding fixed the value of i_t.

The calculation of the of the $n+1$-maturity bond price is now simply

an evaluation of this cgf at $s = 1$:

$$\begin{aligned}
\log q_t^{(n+1)} &= \log E\left[e^{\log m_{t+1} + \log q_{t+1}^{(n)}} \mid i_t\right] \\
&= k_{n+1,t}(1) \\
&= -\lambda^2/2 + A_n + B_n\delta(1-\varphi) + (\varphi B_n - 1)i_t + (\lambda + B_n\sigma)^2/2 \\
&= \underbrace{A_n + B_n[\delta(1-\varphi) + \lambda\sigma] + B_n^2\sigma^2/2}_{=A_{n+1}} + \underbrace{(\varphi B_n - 1)}_{=B_{n+1}} z_t.
\end{aligned}$$

This verifies that our loglinear guess was right. But it also gives us a set of recursive equations that the A's and B's must satisfy:

$$\begin{aligned}
A_{n+1} &= A_n + B_n[\delta(1-\varphi) + \lambda\sigma] + B_n^2\sigma^2/2 \\
B_{n+1} &= \varphi B_n - 1 \;\Rightarrow\; B_n = -(1-\varphi^n)/(1-\varphi).
\end{aligned}$$

Given values for the parameters $(\delta, \varphi, \sigma, \lambda)$ we compute these one at a time, starting with $A_0 = B_0 = 0$, *i.e.*, $\log q_t^{(0)} = 0$ for all i_t. It's easy to do this in Matlab, or even a spreadsheet program.

And recall that these formulas also give us the intercept and slope for yields,

$$y_t^{(n)} = -(A_n/n) - (B_n/n)i_t,$$

forward rates,

$$\begin{aligned}
f_t^{(n)} &= \log q_t^{(n)} - \log q_t^{(n+1)} \\
&= (n+1)y_t^{(n+1)} - ny_t^{(n)} \\
&= (A_n - A_{n+1}) + (B_n - B_{n+1})i_t,
\end{aligned}$$

where

$$\begin{aligned}
B_n - B_{n+1} &= \varphi^n \\
A_n - A_{n+1} &= -B_n[(1-\varphi)\delta + \lambda\sigma] - B_n^2\sigma^2/2 \\
&= (1-\varphi^n)[\delta + \lambda\sigma/(1-\varphi)] - [(1-\varphi^n)^2\sigma^2/(1-\varphi)^2]/2.
\end{aligned}$$

Notice that as $n \to \infty$, the dependence of the forward rate on the current interest rate goes to 0 since $|\varphi| < 1$ and the intercept goes to $\delta + \lambda\sigma/(1-\varphi) - [\sigma^2/(1-\varphi)^2]/2$. In other words, as maturity grows, the model predicts that forward rates get less and less variable (since its dependence on the state variable gets a smaller and smaller weight) and ultimately converge to a constant value.

Parameter values.

We now have formulas for bond prices – and hence yields and forward rates – as functions of the current state – the short rate. Here

we use what we know about forward rates to estimate the model's parameters. So what do we know? We have both time-series and cross-section information. Time-series information includes the autocorrelations of interest rates. Cross-section information includes the mean forward rate curve: that long rates are on average greater than short rates. These seem like basic features of interest rates that we would like any reasonable model to reproduce.

Let's choose parameters that match some of the observed properties of forward rates. Some ballpark numbers are reported in Table 3. The numbers are computed from the quarterly data for US Treasuries over the period 1962-2016 that we looked at earlier. They're reported as annual percentages, something that will show up in our calculations. To get annual rates we multiplied quarterly rates by 4, and to get percentages we multiplied them by 100, so we multiplied quarterly rates by 400 altogether.

n	Forward rate $f_t^{(n-1)}$	Mean	Std Dev	Corr(1)	Corr(4)	Corr(40)
1	$f_t^{(0)} = i_t$	5.1149	3.2874	0.9699	0.8545	0.2965
2	$f_t^{(1)}$	5.2112	3.3073	0.9744	0.8775	0.3074
4	$f_t^{(3)}$	5.4881	3.2392	0.9782	0.8971	0.3245
40	$f_t^{(39)}$	7.0550	2.3683	0.9841	0.9144	0.1445

Table 3: Properties of forward rates. Maturity, n, in quarters.

Now let's put these numbers to work. The time interval for our analysis will be one quarter (3 months), so the short rate is the quarterly rate: $f_t^{(0)} = i_t$ in the table.

- *Mean short rate.* The mean short rate is

$$E(i_t) \quad = \quad \delta,$$

  which we match to the observed mean short rate, which gives us a value of $\delta = (5.1149/400) = 0.0128$. (Note that we have converted the numbers in the table from annual percentages to quarterly interest rates by dividing by 400.)

- *Autocorrelation.* Because i_t is an AR(1), its first-order autocorrelation is φ. So we set φ equal to the observed autocorrelation of the short rate in the data. From the table, we have $\varphi = 0.9699$.

- *Variance.* We also know that for the AR(1) model, the variance of the short rate is $\sigma^2/(1-\varphi^2)$. Using data from the same table gives us

$$\mathrm{Var}(i_t) \quad = \quad (3.2874/400)^2 \quad = \quad \sigma^2/(1-\varphi^2) \quad \Rightarrow \quad \sigma = 0.0020.$$

- *Long forward rates.* Next we have λ, which controls the pricing of interest rate risk and therefore the slope of the forward rate curve. We saw earlier that forward rates are

$$f_t^{(n)} = (1-\varphi^n)[\delta + \lambda\sigma/(1-\varphi)] - [(1-\varphi^n)^2\sigma^2/(1-\varphi)^2]/2 + \varphi^n i_t.$$

Since i_t has a mean of δ, we have[23]

$$E\left[f^{(n)} - i_t\right] \quad = \quad \lambda(1-\varphi^n)\sigma/(1-\varphi) - [(1-\varphi^n)^2\sigma^2/(1-\varphi)^2]/2.$$

We solve this numerically. Given the values we have for φ and σ, choose an arbitrary value for λ and compute the $E\left[f^{(n)} - i_t\right]$. Vary λ until we hit the observed mean forward spread for, say $n = 39$: $(7.0550 - 5.1149)/400$. The value that works is $\lambda = 0.1279$.

[23] The difference between a long forward rate and the short rate, $f^{(n)} - i_t$ is called the *forward spread*. It should not be confused with the *forward premium*, $f_t^{(n)} - E_t i_{t+n}$.

(If any of this seems mysterious, see the Matlab program.)

How well does our model do? There are lots of dimensions we can look at. Let's first calculate the mean of $f_t^{(n)} - i_t$ in the model for our calibrated values of $(\delta, \sigma, \varphi, \lambda)$, then Figure 8 plots them against n and compares that curve to the comparable averages in the data.

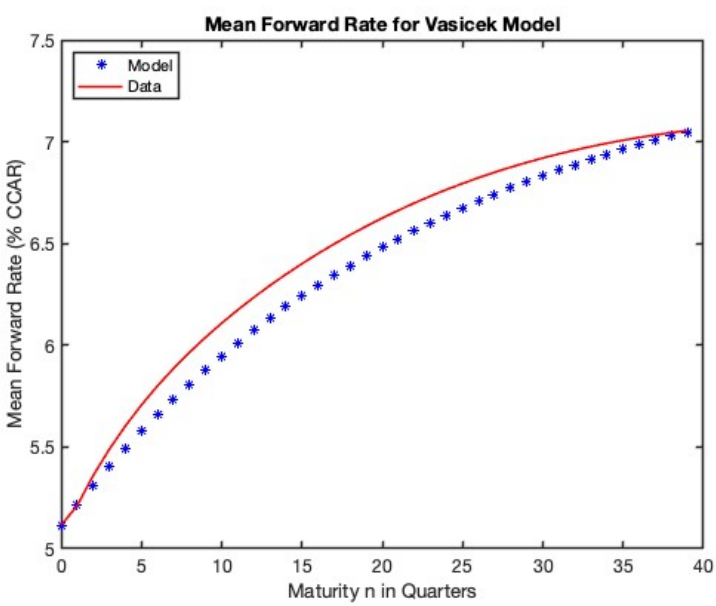

Figure 8: Comparison of predictions of the mean forward-rate curve implied by a calibration of the Vasicek model to average forward rates in a sample from 1962-2016. Parameter values are $\varphi = 0.9699$, $\sigma = 0.0020$, $\lambda = 0.1279$.

Not bad for such a simple model. We get the very short and very long ends of the maturity structure exactly. Yes, of course, we chose parameters to guarantee this. But we don't do too badly for all the other maturities that we didn't use to identify parameters. The model

underpredicts the average spread for maturities in the middle. But for our first attempt, I'd say we got pretty close.

What would happen if we changed the way we identified the price of risk parameter, λ? That is, what if we chose it to fit the the 3-quarter-ahead average forward rate in the data rather than the 39-quarter ahead rate? Vary λ until we hit the observed mean forward spread for, say $n = 3$: $(5.4881 - 5.1149)/400$. The value that works in this case is $\lambda = 0.1631$. The interest rate risk has a substantially larger price.

How do the predictions of the model change? When we graph the mean spreads predicted by the model with this larger price of risk, we see that we do much better matching the data at the short end, but much worse at the long end.

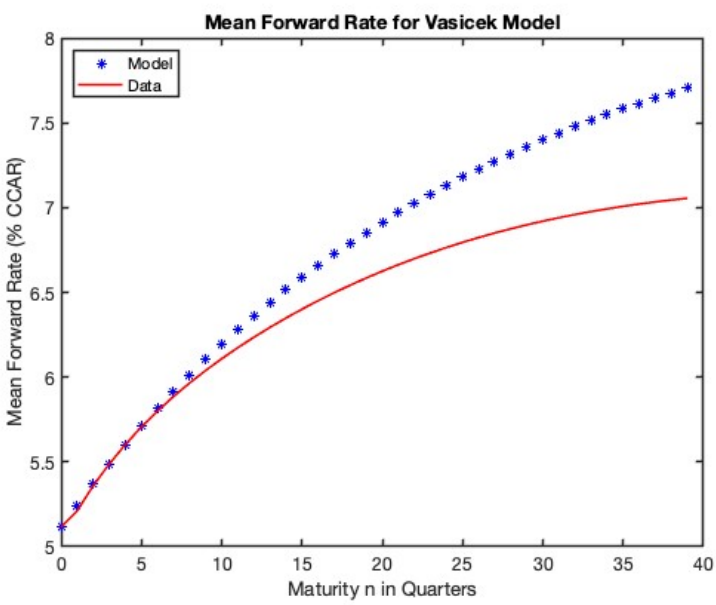

Figure 9: Comparison of predictions of the mean forward-rate curve implied by an alternative calibration of the Vasicek model to average forward rates in a sample from 1962-2016. Parameter values $\varphi = 0.9699$ and $\sigma = 0.0020$ are unchanged, but the price of risk is larger in absolute value, $\lambda = 0.1631$.

The forward curve is steep at the short end implying lots of risk, but then it flattens out suggesting that risk dies out relatively quickly. There seems to be some tension in this model between the persistence of risk, φ, and the price of that risk, λ.

How does the model do capturing the variability of forward rates? The variance in the model is

$$\begin{aligned} \mathrm{Var}\left[f_t^{(n)}\right] &= (B_n - B_{n-1})^2 \mathrm{Var}[i_t] \\ &= \varphi^{2n}\sigma^2/(1-\varphi^2). \end{aligned}$$

In other words, the standard deviation of forward rates declines geometrically as the maturity grows, with the rate of decay given by φ^2. How does that compare with the data?

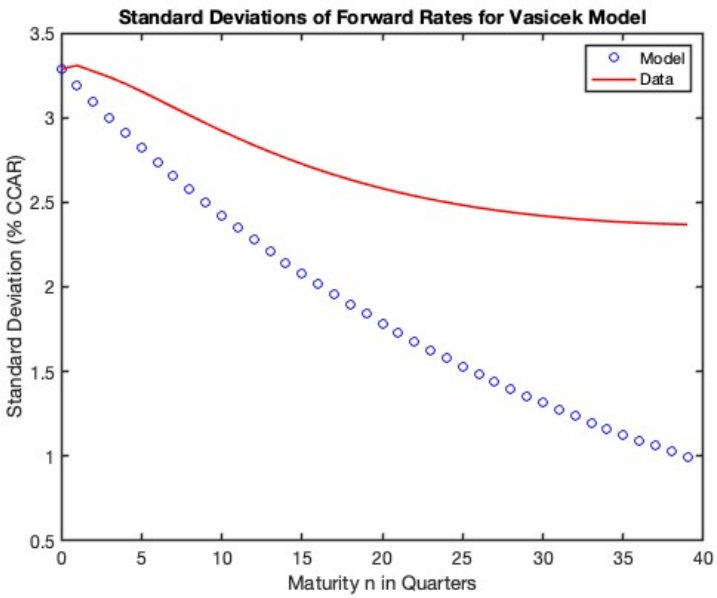

Figure 10: Comparison of predictions of the standard deviation for the forward-rate curve implied by a calibration of the Vasicek model to the standard deviation of forward rates in a sample from 1962-2016. Parameter values are $\varphi = 0.9699$ and $\sigma = 0.0020$.

We can see that we match the volatility of the short rate exactly. No surprise since that was how we identified the parameter σ. It is evident that the model implies substantially less volatility at the long end than we see in the data. We could try to match this by making the value of φ closer to 1, but then short rates would be more persistent in the model than they appear to be in the data. Moreover, we would be amplifying the tension between persistence and the price of risk that we saw in the means.

Overall, our first attempt does surprisingly well, but it seems to be a bit too simple to exactly replicate many of the patterns we seen in the data. We'll have to think a bit on how we can improve the performance of this model by introducing other features into the stochastic process for the pricing kernel.

Digression: risk-neutral probabilities in the Vasicek model

Notice in the formulas for A_n and B_n in the Vasicek model, the price-of-risk parameter, λ, appears in only one spot: the term multiplying B_{n-1} in the recursive formula for A_n. That implies that the only

place that this model differs from a model with a risk-neutral pricing kernel, *i.e.*, $\lambda = 0$, is in that particular term.

How would the bond prices and yields of the Vasicek model differ from a slightly altered risk-neutral version of the model? The altered version we have in mind is

$$\begin{aligned} i_{t+1} &= (1-\varphi)\delta^* + \varphi i_t + \sigma w_{t+1} \\ \log m^*_{t+1} &= -i_t \,, \end{aligned} \tag{5}$$

where we've cleverly chosen δ^*, *the mean of* i_t, to be a function of the original values of δ and λ according to the equation $(1-\varphi)\delta^* = (1-\varphi)\delta + \lambda\sigma$, which implies $\delta^* = \delta + \lambda\sigma/(1-\varphi)$.

The answer is *nothing* changes. The coefficients for A_n and B_n are identical! In other words, we've just identified the *risk-neutral* distribution for the Vasicek model. It has the same persistence and the same variance as the original distribution, *i.e.*, φ and σ are the same in both versions. But the risk-neutral distribution makes an adjustment to the mean of i_t that encodes the price of risk parameter, λ.

Bond pricing with the risk-neutral distribution follows the same steps. Guess a loglinear function, $\log q_t^{(n)} = A_n + B_n i_t$, and apply it to the recursive relationship,

$$\begin{aligned} q_t^{(n+1)} &= E_t\left[m_{t+1} q_t^{(n)}\right] \\ &= e^{-i_t} E_t^*\left[q_t^{(n)}\right]. \end{aligned}$$

Since the distribution for i_{t+1} conditional i_t under the *risk-neutral* distribution is still normal, the conditional cgf for the risk-neutral distribution is

$$\begin{aligned} k_t^*(s) &= \log E_t^*\left[e^{s\, i_{t+1}}\right] \\ &= s[(1-\varphi)\delta^* + \varphi i_t] + (s^2/2)\sigma^2, \end{aligned}$$

where we've used a superscript $*$ to denote the risk-neutral distribution. The conditional cgf under the risk-neutral distribution for the linear function of i_{t+1}, $\log q_{t+1}^{(n)} = A_n + B_n i_{t+1}$ is then

$$\begin{aligned} k^*_{n+1,t}(s) &= \log E_t^*\left[e^{s(A_n + B_n i_{t+1})}\right] \\ &= s[A_n + B_n(1-\varphi)\delta^* + B_n\varphi i_t] + (s^2/2)B_n^2\sigma^2. \end{aligned}$$

Finally, we can find the $n+1$-maturity bond price by evaluating this

cgf at $s = 1$,

$$\begin{aligned}
\log q_t^{(n+1)} &= -i_t + \log E_t^* \left[e^{\log q_t^{(n+1)}} \right] \\
&= -i_t + k_{n+1,t}^*(1) \\
&= -i_t + [A_n + B_n(1-\varphi)\delta^* + B_n \varphi i_t] + B_n^2 \sigma^2 / 2 \\
&= \underbrace{A_n + B_n(1-\varphi)[\delta + \lambda\sigma/(1-\varphi)] + B_n^2\sigma^2/2}_{=A_{n+1}} + (\underbrace{B_n\varphi - 1}_{B_{n+1}}) i_t,
\end{aligned}$$

which, as expected, are the same recursive equations we had before.

Digression: Interest rate risk and duration

The *duration* of a bond is the percentage decrease in its price in response to an increase in its yield to maturity. We know that the prices of pure discount bonds are connected to yields by

$$q_t^{(n)} = e^{-n y_t^{(n)}}$$

so duration is just maturity

$$\begin{aligned}
D &= -\frac{\partial \log q_t^{(n)}}{\partial y_t^{(n)}} = n \\
&\Rightarrow \frac{\partial q_t^{(n)}}{\partial y_t^{(n)}} = -n q_t^{(n)}.
\end{aligned}$$

(This is particularly clean with continuous compounding, but you get something similar with other kinds of compounding.)

There's nothing controversial about this definition until we think about how duration is often used by finance practitioners. They ignore the distinction between a change in $y_t^{(n)}$ and a change in the interest rate, i_t, and act as if duration is the percentage decrease in the price of a long bond in response to an increase in the interest rate. We can use the Vasicek model as a laboratory for thinking about interest rate risk, and whether these kinds of practitioner shortcuts have any real consequence.

In the Vasicek model, a change in $\log q_t^{(n)}$ given a change in short rate, i_t, is determined by B_n, which measures the interest-rate sensitivity of a long bond price:

$$\begin{aligned}
-\frac{\partial \log q_t^{(n)}}{\partial i_t} = -\frac{\partial (A_n + B_n i_t)}{\partial i_t} &= -B_n \\
&= (1-\varphi^n)/(1-\varphi).
\end{aligned}$$

When practitioners behave has if $D = n$ it's as if they're assuming that $B_n = -n$, which is the limiting case of $\varphi = 1$.[24] In other words, they behave as if all changes in interest rates have a permanent effect into the infinite future: $E_t[i_{t+n}] = i_t$ for all n.

[24] Use L'Hôpital's rule to show that $\lim_{\varphi \to 1} B_n = -n$.

In the real world, the yield curve doesn't shift in parallel in response to a change in the interest rates. That's not surprising given what the Vasicek model teaches us: the interest-rate sensitivity of a long bond yield is $-B_n/n$ which is not equal to 1. With mean reversion in interest rates ($0 < \varphi < 1$), sensitivity $-B_n$ increases more slowly than n. We get $B_n = n$ only in the limiting case $\varphi = 1$. Otherwise, mean reversion moderates the sensitivity of long yields to movements in the short rate.

Using duration to measure interest-rate sensitivity of long bonds is a shortcut that has been used for decades, but it's not really the measure of interest rate sensitivity that people often assume. For example, imagine you owned an n-period pure discount bond and wanted to *hedge* its interest-rate risk. (That's just finance jargon for adding another investment to your portfolio to reduce the sensitivity of portfolio's value to changes in the interest rate.) Let's consider an investment in the 1-period bond of an amount h_t, so that the value of my portfolio is now $q_t^{(n)} + h_t q_t^{(1)}$. What value of h_t will make this portfolio insensitive to changes in i_t? The Vasicek model gives us an exact prediction:

$$\begin{aligned}
\frac{\partial\left[q_t^{(n)} + h_t q_t^{(1)}\right]}{\partial i_t} &= \frac{\partial\left[e^{A_n + B_n i_t} + h_t e^{-i_t}\right]}{\partial i_t} \\
&= q_t^{(n)} B_n - h_t q_t^{(1)} = 0 \\
\Rightarrow h_t &= -\left[(1-\varphi^n)/(1-\varphi)\right] q_t^{(n)} / q_t^{(1)}.
\end{aligned}$$

The interpretation of this strategy is simple. When i_t goes up by a small amount, $q_t^{(n)}$ goes down by an amount $B_n q_t^{(n)}$ (remember that B_n is negative in this parameterization). To offset this, sell some of the one-period bond $q_t^{(1)}$, since it's price will also go down when i_t goes up, which creates an offsetting gain.[25] The hedge, h_t, is the amount that makes these gains and losses balance. For $n = 40$ and $\varphi = 0.9699$, mistakenly using duration to calculate this hedge, *i.e.* mistakenly assuming that $B_n = -n$, results in a value of h_t that is 71% too big.

[25] The opposite direction is what often makes hedging difficult to manage in a large organization. Imagine i_t goes down so that $q_t^{(n)}$ goes up. We're long that bond, so we made money – nice! But wait, the hedge part of our portfolio is *designed* to intentionally *lose* money in this case. We are short the 1-period bond, and when i_t goes down, $q_t^{(1)}$ also goes up, so we lose money. Now think about how you will explain to a financially illiterate boss that it's a good thing that you intentionally lost money.

Risk premiums

Another feature of the Vasicek model that we could explore further is the risk premium implicit in long-term bond prices. We know that since the short rate is an AR(1), the conditional expectation at date-t of the short-rate n periods in the future at $t+n$ is

$$E_t(i_{t+n}) \quad = \quad (1-\varphi^n)\delta + \varphi^n\, i_t.$$

Note that the forward rate set today for a one-period forward bond for date $t+n$, can be decomposed into an expected short rate component plus a risk premium:

$$f_t^{(n)} - E_t(i_{t+n}) \quad = \quad \lambda^2/2 - \left[\lambda + \sigma(1-\varphi^n)/(1-\varphi)\right]^2/2.$$

For example, the one-period risk premium imbedded in a 2-period bond is

$$\begin{aligned} f_t^{(1)} - E_t(i_{t+1}) &= \lambda^2/2 - (\lambda+\sigma)^2/2 \\ &= -\sigma^2/2 - \lambda\sigma \\ &= (-1/2 - \lambda/\sigma)\underbrace{\mathrm{Var}_t(i_{t+1})}_{=\sigma^2 \;\; constant}. \end{aligned}$$

That is, the arbitrage-free forward rate includes not just what investors expect the comparable short rate will be in the future, but since they are locking in to a fixed-rate, there's also compensation for the risk of future interest-rate fluctuation. This risk premium depends on both the price of risk, λ, and the quantity of risk, σ.

You might be asking yourself why the risk premium depends on a variance, when we've shown previously that it's the covariance of the pricing kernel with the future cashflow of an asset that determines its riskiness. It turns out that in these models, the conditional covariance is a function of the conditional variance. To see this in the $n = 2$ example, let's expand the basic pricing equation into it's various components, recall that all variables in the model are loglinear in the

normal random variable z_t, use our usual tricks, and rearrange terms:

$$\begin{aligned}
q_t^{(2)} &= E_t\left[m_{t+1}q_{t+1}^{(1)}\right] \\
&= E_t\left[e^{\log m_{t+1}+\log q_{t+1}^{(1)}}\right] \\
&= E_t\left[e^{\log m_{t+1}-i_{t+1}}\right] \\
\Rightarrow \log q_t^{(2)} &= E_t(\log m_{t+1} - i_{t+1}) + \mathrm{Var}_t(\log m_{t+1} - i_{t+1})/2 \\
&= \overbrace{E_t(\log m_{t+1}) + \mathrm{Var}_t(\log m_{t+1})/2}^{=\log q_t^{(1)}} \\
&\quad -E_t(i_{t+1}) + \mathrm{Var}_t(i_{t+1})/2 - \mathrm{Cov}_t(\log m_{t+1}, i_{t+1})
\end{aligned}$$

$$\begin{aligned}
\Rightarrow \overbrace{\log q_t^{(1)} - \log q_t^{(2)}}^{f_t^{(1)}} - E_t(i_{t+1}) &= -\mathrm{Var}_t(i_{t+1})/2 + \mathrm{Cov}_t(\log m_{t+1}, i_{t+1}) \\
&= -\sigma^2/2 + \underbrace{\mathrm{Cov}_t(\lambda w_{t+1}, \sigma w_{t+1})}_{=\lambda\sigma}
\end{aligned}$$

That is, the pricing kernel covaries with the future cashflow, i_{t+1} precisely because they both depend on w_{t+1}. When $\lambda = 0$ we think of the representative agent as *risk neutral*, but the risk premium isn't exactly zero since we still have a $-\sigma^2/2$ term remaining, which is just an artifact of working with log-normal random variables.

It is apparent from the risk premium imbedded in log bond prices that this risk premium varies with maturity. Since $|\varphi| < 1$, it grows with maturity but converges to a constant as n gets large

$$\begin{aligned}
\lim_{n\to\infty}\left[f_t^{(n)} - E_t(i_{t+n})\right] &= \lim_{n\to\infty}\left[\lambda^2/2 - \left[\lambda - \sigma(1-\varphi^n)/(1-\varphi)\right]^2/2\right] \\
&= \lambda^2/2 - [\lambda - \sigma/(1-\varphi)]^2/2.
\end{aligned}$$

It is also apparent that the risk premium for any maturity does not depend on the current state, z_t.

We looked at data for $f_t^{(n)} - i_t$ (often referred to as the *forward spread* to distinguish it from the *risk premium*), which isn't quite what we're looking for.[26] But since it was moving around over time, we'll make the leap and conclude that the risk premium is also moving around over time. If we wanted to spend a bit more time, we could come up with many ways to demonstrate that the risk premiums evident in the data we see are *not* constant across time. This motivates our next class of models in which the conditional variance of the pricing kernel depends on the state variable, z_t, which will ultimately deliver risk premiums that also move around over time.

[26] Note that the risk premium, $f_t^{(n)} - E_t(i_{t+n})$ is equal to the forward spread $f_t^{(n)} - i_t$ minus the error you would make forecasting the future short rate with the current short rate, $E_t(i_{t+n}) - i_t$. As we've seen, the short rate is highly persistent, so the second piece is approximately zero. Therefore, we don't lose much information using the forward spread as a proxy for the risk premium.

The Cox-Ingersoll-Ross model

Another simple model within the general class of exponential-affine models that *can* generate risk premium that vary with the state, is something called the Cox-Ingersoll-Ross model.[27] Let's see how it works.

[27] Cox, J.C., J.E. Ingersoll and S.A. Ross, 1985. "A Theory of the Term Structure of Interest Rates," *Econometrica* 53, 385–407. Note that the derivation of this model is also done in continuous time, but like the Vasicek model, it translates nicely to our discrete-time framework.

Model. We start with a variant of the Vasicek model but we allow for risk to also vary with the state z_t. (More on that shortly.) The model is

$$\begin{aligned} \log m_{t+1} &= -(1+\lambda^2/2)z_t + \lambda z_t^{1/2} w_{t+1} \\ z_{t+1} &= (1-\varphi)\delta + \varphi z_t + \sigma z_t^{1/2} w_{t+1}, \end{aligned} \tag{6}$$

with (as usual) independent standard normal innovations w_t. We've once again chosen a parameterization that may look a little odd at this point, but it is without loss of generality, and it will once again result in the convenient result connecting z_t directly to the short rate.

Here z_t plays two roles: It governs the conditional mean of the pricing kernel, as it did before, but it also governs the conditional variance:

$$\begin{aligned} E_t(z_{t+1}) &= (1-\varphi)\delta + \varphi z_t \\ \mathrm{Var}_t(z_{t+1}) &= \sigma^2 z_t. \end{aligned}$$

And since conditional on z_t, z_{t+1} is still a linear function of a normal random variable, the conditional distribution is also normal,

$$\log z_{t+1} \mid z_t \sim \mathcal{N}\big((1-\varphi)\delta + \varphi z_t\,,\ \sigma^2 z_t\big).$$

The conditional cgf for z_{t+1}, conditional on z_t, is

$$\begin{aligned} k_{zt}(s) &= \log E\big[e^{sz_{t+1}} \mid z_t\big] \\ &= s[(1-\varphi)\delta + \varphi z_t] + (s^2/2)\sigma^2 z_t. \end{aligned}$$

Likewise, conditional on z_t, the log of the pricing kernel is also a linear function of a normal random variable, so it's conditional distribution is also normal,

$$\log m_{t+1} \mid z_t \sim \mathcal{N}\big(-(1+\lambda^2/2)z_t\,,\ \lambda^2 z_t\big).$$

Note once again, z_t determines *both* the conditional mean and the conditional variance of $\log m_{t+1}$.[28] The conditional cgf for $\log m_{t+1}$,

[28] Hint: Since we saw in the Vasicek model that the risk premium depended on the conditional variance of $\log m_{t+1}$, making that conditional variance a function of z_t will likely make the risk premium a function of z_t too.

conditional on z_t, is

$$\begin{aligned} k_{mt}(s) &= \log E\left[e^{s \log m_{t+1}} \mid z_t\right] \\ &= s[-(1+\lambda^2/2)z_t] + (s^2/2)\lambda^2 z_t. \end{aligned}$$

In the AR(1) model, z_t could take on positive or negative values since w_t was a standard normal. Shouldn't we be worried that making the conditional variance proportional to z_t could give us a nonsensical model with negative variances? The so-called "square-root" process (6) guarantees that it stays positive – at least over short intervals of time. The idea is that the variance shrinks as z_t gets close to zero. As a result, the change in z_t near zero comes from the constant term,

$$z_{t+1} - z_t \approx (1-\varphi)\delta > 0.$$

As long as $\delta > 0$, that pushes z_t away from zero and keeps it positive.

Aside. If you're concerned about the positive correlation built into the conditional mean and conditional variance – big values of z_t means the conditional distribution of z_{t+1} will necessary have a big mean when $\phi > 0$ and a big variance – there's no need to worry. A small generalization takes care of this without adding to our analytical burden: $z_{t+1} = (1-\varphi)\delta + \varphi z_t + [\gamma_0 + \gamma_1 z_t]^{1/2} w_{t+1}$. We can scale γ_0 and γ_1 to insure a positive variance with virtual certainty, and γ_1 can be chosen to have the opposite sign of φ, so the conditional mean and conditional variance can move in the opposite directions. Since the model has one more parameter, we'll think about using it only if we need to.[29]

[29] And if we really wanted to get fancy, we could work with a model called an *Autoregressive Gamma* or ARG. You can think of this as like the autoregressive models we've been working with, but instead of the state variable being conditionally normally distributed, it would have a Gamma distribution defined over the non-negative real line. The good news is that this model ends up looking just like the one we work with, so even though we've ignoring some technical issues, we're not really making any big mistakes.

Despite the square-root process, z_t is still an AR(1). Its mean is

$$E(z) = (1-\varphi)\delta + \varphi E(z) \quad \Rightarrow \quad E(z) = \delta.$$

The variance is a bit more complicated that usual because the conditional variance is now also a stochastic process:

$$\begin{aligned} \mathrm{Var}(z) &= \mathrm{Var}\left[E_t(z_{t+1})\right] + E\left[\mathrm{Var}_t(z_{t+1})\right] \\ &= \mathrm{Var}\left[(1-\varphi)\delta + \varphi z_t\right] + E\left[\sigma^2 z_t\right] \\ &= \varphi^2 \mathrm{Var}(z_t) + \sigma^2 \delta \\ \Rightarrow \mathrm{Var}(z) &= \sigma^2 \delta/(1-\varphi^2). \end{aligned}$$

What about autocorrelations? Let's first find the first-order autoco-

variance:

$$\begin{aligned}
\mathrm{Cov}(z_t, z_{t-1}) &= E[(z_t - \delta)(z_{t-1} - \delta)] \\
&= E\left\{[\varphi(z_{t-1} - \delta) + \sigma z_{t-1}^{1/2} w_t](z_{t-1} - \delta)\right\} \\
&= E\big[\varphi(z_{t-1} - \delta)^2 - \delta\varphi(z_{t-1} - \delta) + \sigma z_{t-1}^{1/2}(z_{t-1} - \delta) w_t\big] \\
&= \varphi \underbrace{E\big[(z_{t-1} - \delta)^2\big]}_{=\sigma^2} - \delta\varphi \underbrace{E(z_{t-1} - \delta)}_{=0} + E\big[\sigma z_{t-1}^{1/2}(z_{t-1} - \delta) \underbrace{E_t(w_t)}_{=0}\big] \\
&= \varphi\sigma^2.
\end{aligned}$$

The first-order autocorrelation is therefore

$$\mathrm{Corr}(z_t, z_{t-1}) = \mathrm{Cov}(z_t, z_{t-1})/\mathrm{Var}(z_t) = \varphi.$$

The last term in the second last line of the autocovariance derivation is an application of the law of iterated expectations (yes, this is a very handy property to know when working with stochastic processes).[30] Since it generalizes to any lag length, that same property, namely $E[z_t] = E\big[E_{t-j}(z_t)\big]$, can be applied to show that the autocorrelation of any order of lag is

$$\mathrm{Corr}(z_t, z_{t-j}) = \varphi^j.$$

In other words, the autocorrelations of this model are identical to the AR(1) with a constant conditional variance.

[30] Specifically, the law of iterated expectations implies that $E\big[\sigma z_{t-1}^{1/2}(z_{t-1} - \delta) w_t\big]$ is equal to $E\big(E_t\big[\sigma z_{t-1}^{1/2}(z_{t-1} - \delta) w_t\big]\big)$ which simplifies to $E\big[\sigma z_{t-1}^{1/2}(z_{t-1} - \delta) E_t(w_t)\big]$.

Recursions. We follow the same logic as in the Vasicek model, starting with the short rate

$$\begin{aligned}
i_t = y_t^{(1)} &= -\log E_t\big[e^{\log m_{t+1}}\big] \\
&= -k_{mt}(1) \\
&= (1 + \lambda^2/2) z_t - \lambda^2 z_t/2 \\
&= z_t.
\end{aligned}$$

The short rate is therefore simply z_t. That's why we started with a somewhat odd-looking coefficient multiplying z_t in the pricing kernel – so we would get this result.

For bonds of arbitrary maturity, we use the guess

$$\log q_t^{(n)} = A_n + B_n z_t.$$

That gives us

$$\log m_{t+1} + \log q_{t+1}^{(n)} = [A_n + (1 - \varphi)\delta + [\varphi B_n - (1 + \lambda^2/2)] z_t + (\lambda + B_n \sigma) z_t^{1/2} w_{t+1}.$$

Conditional on z_t, this is just a linear function of a normal random variable, therefore, it too has a conditional normal distribution,

$$(\log m_{t+1} + \log q_{t+1}^{(n)}) \mid z_t \sim \mathcal{N}\big(A_n + (1 - \varphi)\delta + [\varphi B_n - (1 + \lambda^2/2)] z_t\,,\, (\lambda + B_n \sigma)^2 z_t\big),$$

and a conditional cgf of

$$k_{n+1t}(s) \quad = \quad s\big[A_n + (1-\varphi)\delta + [\varphi B_n - (1+\lambda^2/2)]z_t\big] + (s^2/2)(\lambda + B_n\sigma)^2 z_t.$$

This leads immediately to the price of an $n+1$-maturity bond,

$$\begin{aligned}
\log q_t^{(n+1)} &= \log E_t\big[e^{\log m_{t+1} + \log q_{t+1}^{(n)}}\big] \\
&= k_{n+1t}(1) \\
&= A_n + B_n(1-\varphi)\delta + [\varphi B_n - (1+\lambda^2/2)]z_t + \big[(\lambda + B_n\sigma)^2/2\big]z_t \\
&= \underbrace{A_n + B_n(1-\varphi)\delta}_{=A_{n+1}} + \big[\underbrace{\varphi B_n - (1+\lambda^2/2) + (\lambda + B_n\sigma)^2/2}_{=B_{n+1}}\big]z_t.
\end{aligned}$$

Lining up terms, we have

$$\begin{aligned}
A_{n+1} &= A_n + B_n(1-\varphi)\delta \\
B_{n+1} &= \varphi B_n - (1+\lambda^2/2) + (\lambda + B_n\sigma)^2/2 \\
&= B_n(\varphi + \lambda\sigma) - 1 + B_n^2\sigma^2/2
\end{aligned}$$

With initial conditions $A_0 = B_0 = 0$ as before. [Remind yourself why.]

These are a slightly more complicated expressions than the Vasicek model, so we won't try to solve these by hand. But it's just as easy to program, which is what we'll do. But there are a couple of obvious things that are now different. The A_n term looks simpler. The B_n term, however, looks more complicated. And now the price-of-risk parameter, λ, affects the slope coefficient, B_n, directly. (It affects the intercept, A_n, too but only indirectly through B_n.)

Risk-neutral distribution. Notice that the only place the price-of-risk parameter λ appears directly in these recursions is in the equation for B_{n+1} and it appears there in a term combined with the persistence parameter $\varphi + \lambda\sigma$. If we had started with a risk-neutral pricing kernel, *i.e.*, $\lambda = 0$, and a different value for the persistence parameter, say $\varphi^* = \varphi + \lambda\sigma$, then the solution for B_{n+1} would be the same. But since φ also appears in the recursion for A_{n+1}, we would need to make an adjustment there as well. If we also chose a different value for δ, the mean of z_t, say $(1-\varphi^*)\delta^* = (1-\varphi)\delta$, then the solution or A_{n+1} would be the same. In other words, bond prices in this model are equivalent to the bond prices of a risk-neutral pricing kernel but with an adjusted square-root process for z_t:

$$\begin{aligned}
z_{t+1} &= (1-\varphi^*)\delta^* + \varphi^* z_t + \sigma z_t^{1/2} w_{t+1} \qquad (7) \\
\log m_{t+1} &= -i_t,
\end{aligned}$$

where $\varphi^* = \varphi + \lambda\sigma$ and $\delta^* = (1-\varphi)\delta/(1-\varphi^*)$.

Note that similar to the risk-neutral distribution for the Vasicek model in equation (5), the process for the conditional variance is the same in the risk-neutral distribution as in the actual distribution for z_t. But unlike the Vasicek model, the persistence parameter in the risk-neutral distribution for the CIR model is now substantially different. We can anticipate that this might help us match the observed behavior in the data. In essence this model creates a wedge between the persistence of the short rate, φ, and the persistence that would seem to drive the behavior of long-maturity bonds, now φ^*. Since this was a source of tension in the Vasicek model, this is likely to improve the ability of the model to match more of the observed facts.

Time-varying risk premiums. It's worth working out the A_n and B_n up to $n = 2$, so that we can get an expression for the one-period risk premium:

$$\begin{aligned} f_t^{(1)} &= (1-\varphi)\delta + \varphi z_t + (\lambda/\sigma - 1/2)\sigma^2 z_t \\ \Rightarrow f_t^{(1)} - E_t(i_{t+1}) &= (\lambda/\sigma - 1/2)\sigma^2 z_t \\ &= (\lambda/\sigma - 1/2)\underbrace{\mathrm{Var}_t(z_{t+1})}_{\textit{not constant}}, \end{aligned}$$

which is the same form we had for the risk premium in the Vasicek model, except now the risk premium changes with the state z_t.

Parameter Values.

Now let's estimate/calibrate parameters of the model from the properties of forward rates reported in Table 3.

- *Mean.* We set δ equal to the mean short rate: $\delta = 5.1149/400 = 0.0128$.
- *Autocorrelation.* We set φ equal to the observed autocorrelation of the short rate in the data, just as we did before. From the table, that gives us $\varphi = 0.9699$.
- *Variance.* Similarly, the variance of the short rate is the variance of z_t, which is $\sigma^2\delta/(1-\varphi^2)$. Given values for φ and δ, the variance of $(3.2874/400)^2$ in the data implies $\sigma = 0.0177$.
- *Long forward rates.* Again, λ controls the mean forward rate curve. Recall that forward rates are

  $$f_t^{(n)} = \log(q_t^{(n)}/q_t^{(n+1)}) = (A_n - A_{n+1}) + (B_n - B_{n+1})i_t.$$

  Since i_t has a mean of δ, we have

  $$E\left[f_t^{(n)} - i_t\right] = (A_n - A_{n+1}) + (B_n - B_{n+1})\delta - \delta.$$

For $n = 39$, *i.e.*, the long forward rate, we would search for a value for λ that makes $E\left[f_t^{(n)} - i_t\right]$ equal to $(7.0550 - 5.1149)/400$. These are equal for the value $\lambda = 0.9918$. (See the Matlab program for details in solving this nonlinear equation in λ.)

We can see for the next two graphs that this model does just about the same as the Vasicek model in matching the properties of the mean forward curve. But where it does substantially better is in matching the volatility of forward rates. In the CIR model, time-varying risks make the forward rates out to 20 quarters roughly as volatile as we see in the data, which is a much better outcome than the constant-risk Vasicek model. On the other hand, the very long end is still more volatile in the data and in the CIR model.

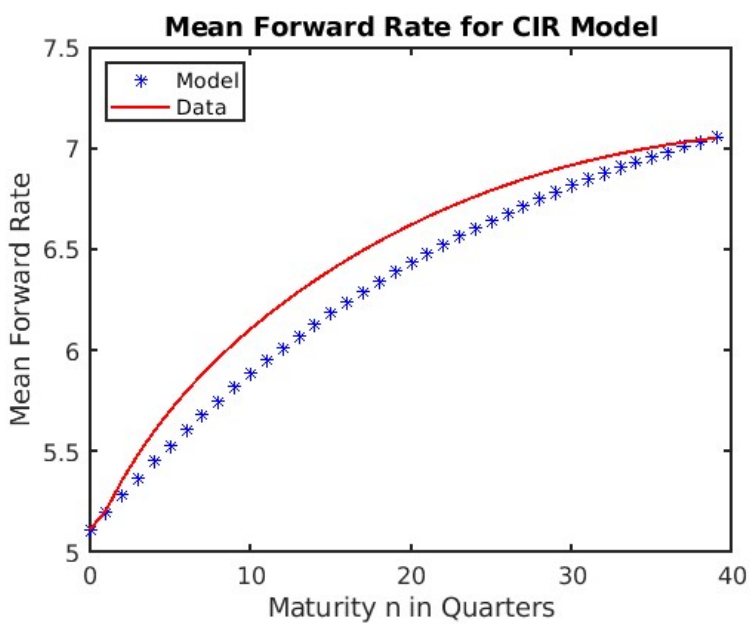

Figure 11: Comparison of predictions of the mean forward-rate curve implied by a calibration of the CIR model to average forward rates in a sample from 1962-2016. Parameter values are $\varphi = 0.9699$, $\sigma = 0.0177$, $\delta = 0.0128$, and $\lambda = 0.9918$.

Is that it? Or are there some relatively easy changes we can make to this model to make it look more like the observed data? In the next section, we'll ask how the addition of a second state variable, *i.e.*, a second source for risk, changes the model's predictions. This will make the algebra a bit more tedious, but no more difficult. And when we see the improved performance, the extra algebra will be time well-spent.

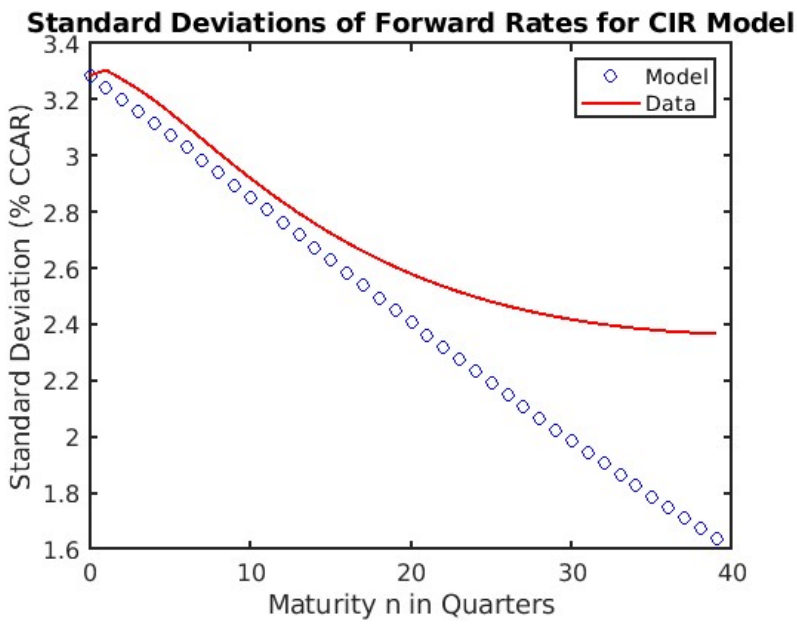

Figure 12: Comparison of predictions of the standard deviation for the forward-rate curve implied by a calibration of the CIR model to the standard deviation of forward rates in a sample from 1962-2016. Parameter values are $\varphi = 0.9699$, $\sigma = 0.0177$, $\delta = 0.0128$, and $\lambda = 0.9918$.

Multi-factor models

When thinking about real-world applications of these simple models, it seems rather simplistic to assume that there is only one source of risk that affects the pricing kernel. But it turns out that without much extra effort, the analysis we've developed extends to as many risk factors as we like. We'll stop at 2-factor models. But you should be able to guess how the extension to 3 or more risk factors will work.

I. Multi-factor Vasicek models

$$\begin{aligned} z_{1t+1} &= (1-\varphi_1)\delta_1 + \varphi_1 z_{1t} + \sigma_1 w_{1t+1} \\ z_{2t+1} &= (1-\varphi_2)\delta_2 + \varphi_2 z_{2t} + \sigma_2 w_{2t+1} \end{aligned}$$

$$\log m_{t+1} = -\lambda_1^2/2 - \lambda_2^2/2 - z_{1t} - z_{2t} + \lambda_1 w_{1t+1} + \lambda_2 w_{2t+1},$$

where both w_{1t} and w_{2t} are *iid* normals – independent across time and independent of each other.[31] Notice that compared to the 1-factor model we now have four additional parameters: λ_2, φ_2, δ_2, and σ_2.

[31] We could make this more complicated by building some correlation across the w's, or making the conditional mean of z_1 depend on the lagged value of the z_2 or vice versa. Same algebra... many more parameters.

The conditional distribution for m_{t+1} conditional on z_{1t} and z_{2t} is

$$\log m_{t+1} \mid z_{1t}, z_{2t} \sim \mathcal{N}\big(-\lambda_1^2/2 - \lambda_2^2/2 - z_{1t} - z_{2t}, \lambda_1^2 + \lambda_2^2\big),$$

and the conditional cgf for m_{t+1} conditional on z_{1t} and z_{2t} is then

$$\begin{aligned} k_{mt}(s) &= \log E\left[e^{s \log m_{t+1}} \mid z_{1t}, z_{2t}\right] \\ &= s(-\lambda_1^2/2 - \lambda_2^2/2 - z_{1t} - z_{2t}) + (s^2/2)(\lambda_1^2 + \lambda_2^2). \end{aligned}$$

We would follow the same method as for the 1-factor Vasicek model but applied to both factors. Begin with the short rate,

$$\begin{aligned} i_t &= -k_{mt}(1) \\ &= z_{1t} + z_{2t}. \end{aligned}$$

Notice that in this case, we are allowing the short rate to depend on both sources of risk.[32]

[32] Later we will link these sources of risk to macroeconomic fundamentals like consumption growth and the inflation rate.

Once again, we assume a loglinear structure for yields:

$$\log q_t^{(n)} = A_n + B_{1n} z_{1t} + B_{2n} z_{2t}$$

and use our familiar tricks to find A_n, B_{1n}, and B_{2n} as recursive functions of n and the other parameters of the model. Consider the random variable $\log m_{t+1} + \log q_{t+1}^{(n)}$, and notice that it is a linear function of conditionally normal random variables, hence, it too is conditionally normal,

$$\begin{aligned} \log m_{t+1} + \log q_{t+1}^{(n)} &= -\lambda_1^2/2 - \lambda_2^2/2 - z_{1t} - z_{2t} + \lambda_1 w_{1t+1} + \lambda_2 w_{2t+1} + A_n + B_{1n} z_{1t+1} + B_{2n} z_{2t+1} \\ &= A_n - \lambda_1^2/2 - \lambda_2^2/2 + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\ &\quad + (B_{1n}\varphi_1 - 1) z_{1t} + (B_{2n}\varphi_2 - 1) z_{2t} \\ &\quad + (\lambda_1 + B_{1n}\sigma_1) w_{1t+1} + (\lambda_2 + B_{2n}\sigma_2) w_{2t+1}, \end{aligned}$$

hence,

$$\begin{aligned} \left(\log m_{t+1} + \log q_{t+1}^{(n)}\right) \mid z_{1t}, z_{2t} \sim \mathcal{N}\Big(& A_n - \lambda_1^2/2 - \lambda_2^2/2 + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\ & + (B_{1n}\varphi_1 - 1) z_{1t} + (B_{2n}\varphi_2 - 1) z_{2t}, \\ & (\lambda_1 + B_{1n}\sigma_1)^2 + (\lambda_2 + B_{2n}\sigma_2)^2 \Big). \end{aligned}$$

This implies a cumulant generating function conditional on z_{1t} and z_{2t},

$$\begin{aligned} k_{n+1t}(s) &= \log E\left[e^{s(\log m_{t+1} + \log q_{t+1}^{(n)})} \mid z_{1t}, z_{2t}\right] \\ &= s\big[A_n - \lambda_1^2/2 - \lambda_2^2/2 + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\ &\qquad + (B_{1n}\varphi_1 - 1) z_{1t} + (B_{2n}\varphi_2 - 1) z_{2t}\big] \\ &\quad + (s^2/2)\left[(\lambda_1 + B_{1n}\sigma_1)^2 + (\lambda_2 + B_{2n}\sigma_2)^2\right], \end{aligned}$$

which we then use to find the price of an $n+1$-maturity bond,

$$\begin{aligned}
\log q_t^{(n+1)} &= \log E_t\left[e^{\log m_{t+1}+\log q_{t+1}^{(n)}}\right] \\
&= k_{n+1t}(1) \\
&= -\lambda_1^2/2 - \lambda_2^2/2 + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\
&\quad +(\lambda_1 + B_{1n}\sigma_1)^2/2 + (\lambda_2 + B_{2n}\sigma_2)^2/2 \\
&\quad +(B_{1n}\varphi_1 - 1)z_{1t} + (B_{2n}\varphi_2 - 1)z_{2t}.
\end{aligned}$$

Line up terms to find the recursive solutions for the intercept and slopes of the log bond price:

$$\begin{aligned}
B_{1n+1} &= -(1-B_{1n}\varphi_1) \;\Rightarrow\; B_{1n} = -(1-\varphi_1^n)/(1-\varphi_1) \\
B_{2n+1} &= -(1-B_{2n}\varphi_2) \;\Rightarrow\; B_{2n} = -(1-\varphi_2^n)/(1-\varphi_2) \\
A_{n+1} &= A_n + B_{1n}[(1-\varphi_1)\delta_1 + \lambda_1\sigma_1] + B_{2n}[(1-\varphi_2)\delta_2 + \lambda_2\sigma_2] \\
&\quad +B_{1n}^2\sigma_1^2/2 + B_{2n}^2\sigma_2^2/2.
\end{aligned}$$

You should notice the similarity with the solution to the 1-factor Vasicek model. In fact, we probably could've guessed this solution based on how we know these models work.

Risk-neutral distribution. Note once again that the only place the two price-of-risk parameters, λ_1 and λ_2, are in the intercept term. As in the 1-factor model, the risk-neutral distributions for z_{1t} and z_{2t} for the 2-factor model are quite simple:

$$\begin{aligned}
z_{1t+1} &= (1-\varphi_1)\delta_1^* + \varphi_1 z_{1t} + \sigma_1 w_{1t+1} \\
z_{2t+1} &= (1-\varphi_2)\delta_2^* + \varphi_2 z_{2t} + \sigma_2 w_{2t+1} \\
\log m_{t+1}^* &= -i_t,
\end{aligned}$$

where $\delta_1^* = \delta_1 + \lambda_1\sigma_1/(1-\varphi_1)$ and $\delta_2^* = \delta_2 + \lambda_2\sigma_2/(1-\varphi_2)$. That is, the bond prices in this model look just like they came from a risk-neutral investor who used a pricing kernel $\log m_{t+1}^* = -i_t$ and adjusted mean parameters in the two AR(1) processes for the two state variables.

And recall that we can also solve for the intercept and slopes for forward rates,

$$f_t^{(n)} = \log q_t^{(n)} - \log q_t^{(n+1)} = (A_n - A_{n+1}) + (B_{1n} - B_{1n+1})z_{1t} + (B_{2n} - B_{2n+1})z_{2t},$$

where

$$\begin{aligned}
B_{1n} - B_{1n+1} &= \varphi_1^n \\
B_{2n} - B_{2n+1} &= \varphi_2^n \\
A_n - A_{n+1} &= (1-\varphi_1^n)[\delta_1 + \lambda_1\sigma_1/(1-\varphi_1)] + (1-\varphi_2^n)[\delta_2 + \lambda_2\sigma_2/(1-\varphi_2)] \\
&\quad -[(1-\varphi_1^n)^2\sigma_1^2/(1-\varphi_1)^2]/2 - [(1-\varphi_2^n)^2\sigma_2^2/(1-\varphi_2)^2]/2.
\end{aligned}$$

Notice that once again as $n \to \infty$, the slope coefficients are both going to 0 since $|\varphi_1| < 1$ and $|\varphi_2| < 1$. The intercept goes to $\delta_1^* + \delta_2^* - [\sigma_1/(1-\varphi_1)]^2/2 - [\sigma_2/(1-\varphi_2)]^2/2$. In other words, as maturity grows, once again long forward rates are becoming less and less variable and ultimately converging to a constant value, just like in the 1-factor Vasicek model.

Will the addition of this state variable improve the empirical performance of this model? On some dimensions, yes.[33] But on other dimensions, the answer, unfortunately, will be no. This 2-factor version of the Vasicek model shares a particularly problematic feature of the 1-factor Vasicek model: the model predicts that risk premiums are constant when we are fairly certain that they are quite variable in the data. To see this, calculate the risk premium on a 1-period forward rate:

$$f_t^{(1)} - E_t(i_{t+1}) = (\lambda_1/\sigma_1 - 1/2)\sigma_1^2 + (\lambda_2/\sigma_2 - 1/2)\sigma_2^2.$$

And risk premiums on longer-maturity forward rates are also constant.

[33] Remember that it can't do any worse, of course, since the 1-factor model is a special case of this model (*i.e.*, set all of the parameters with a subscript of 2 equal to zero, and you get back the 1-factor model).

We could do an empirical example of this two-factor model as we did with the one-factor Vasicek model. What we would learn is that the second factor will help us better match the curvature in the average forward curve. But the model would still predict volatilities at the long that are much much smaller than we see in the data. Moreover, we know it predicts constant risk premiums which looks nothing like the data. This motivates us to look at a CIR model with multiple factors, which we then fit to data.

II. Multi-factor CIR models

What we learn from these quantitative exercises is that we will likely need more than one source of time-varying risk to capture the most important features of the term structure. The mean forward curves is telling us that there are some high-priced risks that are relatively short-lived and create a steep forward curve at the short end. In addition, there seem to be some more persistent risks that have a lower price, but which dominate at longer maturities. And the slowing decaying volatilities of forward rates and other evidence on time-varying risk premiums, suggest that these risks may all have time-varying conditional variances.

We now turn extend the CIR model to incorporate these features. The model is

$$\begin{aligned}
\log m_{t+1} &= -z_{1t} - (\lambda_1^2/2 + \lambda_2^2/2) z_{2t} \\
&\quad + \lambda_1 z_{2t}^{1/2} w_{1t+1} + \lambda_2 z_{2t}^{1/2} w_{2t+1} \\
z_{1t+1} &= (1-\varphi_1)\delta_1 + \varphi_1 z_{1t} + z_{2t}^{1/2} w_{1t+1} \\
z_{2t+1} &= (1-\varphi_2)\delta_2 + \varphi_2 z_{2t} + \sigma_2 z_{2t}^{1/2} w_{2t+1},
\end{aligned} \tag{8}$$

where w_{1t} and w_{2t} are still iid N(0,1), and they are also independent of each other at all leads and lags. Notice that $z_{2t}^{1/2}$ is scaling both shocks. In that sense, we can think of z_{2t} as a stochastic process that determines the conditional variance. There are now many more parameters: two prices of risk, λ_1 and λ_2, three parameters for the variance process z_{2t}, $(\delta_2, \varphi_2, \sigma_2)$, and two parameters for the process for z_{1t}, (δ_1, φ_1). This is referred to as a 2-factor model for obvious reasons. You will see that the logic and the solution method extends to any number of factors. How many factors, therefore, will be determined by the performance of the model.

The conditional distribution for z_{1t+1} and z_{2t+1} are similar to the 1-factor CIR model:

$$\begin{aligned}
z_{1t+1} \mid z_{1t}, z_{2t} &\sim \mathcal{N}\big((1-\varphi_1)\delta_1 + \varphi_1 z_{1t}, z_{2t}\big) \\
z_{2t+1} \mid z_{1t}, z_{2t} &\sim \mathcal{N}\big((1-\varphi_2)\delta_2 + \varphi_2 z_{2t}, \sigma_2^2 z_{2t}\big),
\end{aligned}$$

so that the conditional cgfs are

$$\begin{aligned}
k_{1t} &= s\big[(1-\varphi_1)\delta_1 + \varphi_1 z_{1t}\big] + (s^2/2) z_{2t} \\
k_{2t} &= s\big[(1-\varphi_2)\delta_2 + \varphi_2 z_{2t}\big] + (s^2/2)\sigma_2^2 z_{2t}.
\end{aligned}$$

Since conditional on values for z_{1t} and z_{2t}, the log pricing kernel is a linear function of normal random variables, it too is a normal random variable:

$$\log m_{1t+1} \mid z_{1t}, z_{2t} \sim \mathcal{N}\big(-z_{1t} - (\lambda_1^2/2 + \lambda_2^2/2) z_{2t}, (\lambda_1^2 + \lambda_2^2) z_{2t}\big),$$

so its conditional cgf is

$$\begin{aligned}
k_{mt}(s) &= \log E\big[e^{s \log m_{t+1}} \mid z_{1t}, z_{2t}\big] \\
&= s\big[-z_{1t} - (\lambda_1^2/2 + \lambda_2^2/2) z_{2t}\big] + (s^2/2)(\lambda_1^2 + \lambda_2^2) z_{2t}.
\end{aligned}$$

We follow the same logic as in the CIR model, starting with the short rate

$$\begin{aligned}
i_t = y_t^{(1)} &= -\log E_t\big[e^{\log m_{t+1}}\big] \\
&= -k_{mt}(1) \\
&= z_{1t} + (\lambda_1^2/2 + \lambda_2^2/2) z_{2t} - (\lambda_1^2 + \lambda_2^2) z_{2t}/2 \\
&= z_{1t}.
\end{aligned}$$

In other words, we've scaled the pricing kernel so that the first factor is the short rate, and the second factor is the conditional variance of the short rate:

$$\mathrm{Var}_t(i_{t+1}) = z_{2t}.$$

The unconditional moments of the short rate are:

$$\begin{aligned} E(i_t) &= \delta_1 \\ \mathrm{Var}(i_t) &= \delta_2/(1-\varphi_1^2). \end{aligned}$$

The autocorrelations are the same as for the AR(1) and 1-factor CIR models

$$\mathrm{Corr}(i_t, i_{t-j}) = \varphi_1^j.$$

Equilibrium bond prices for longer maturities follow from the same recursive, arbitrage-free arguments that should by now feel routine. We guess a log linear function for the bond price, then solve recursively. The only thing that has changed is that one of our two state variables is a conditional variance. Our guess – which is becoming almost automatic – is

$$\log q_t^{(n)} = A_n + B_{1n} z_{1t} + B_{2n} z_{2t},$$

and of particular interest is the sum of $\log m_{t+1}$ and $\log q_{t+1}^{(n)}$,

$$\begin{aligned} \log m_{t+1} + \log q_{t+1}^{(n)} &= -z_{1t} - (\lambda_1^2/2 + \lambda_2^2/2) z_{2t} + \lambda_1 z_{2t}^{1/2} w_{1t+1} + \lambda_2 z_{2t}^{1/2} w_{2t+1} \\ &\quad + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{1n}\varphi_2 z_{1t} + B_{1n} z_{2t}^{1/2} w_{1t+1} \\ &\quad + B_{2n}(1-\varphi_2)\delta_2 + B_{2n}\varphi_2 z_{2t} + B_{2n}\sigma_2 z_{2t}^{1/2} w_{2t+1} \\ &= A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\ &\quad + (B_{1n}\varphi_1 - 1) z_{1t} + \left[B_{2n}\varphi_2 - (\lambda_1^2/2 + \lambda_2^2/2)\right] z_{2t} \\ &\quad + (\lambda_1 + B_{1n}) z_{2t}^{1/2} w_{1t+1} + (\lambda_2 + \sigma_2 B_{2n}) z_{2t}^{1/2} w_{2t+1}. \end{aligned}$$

Since, conditional on z_{1t} and z_{2t}, this is a linear combination of normal random variables, it's conditional distribution is also normal,

$$\begin{aligned} \left(\log m_{t+1} + \log q_{t+1}^{(n)}\right) \mid z_{1t}, z_{2t} \quad \sim \quad & \mathcal{N}\big(A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\ & \quad + (B_{1n}\varphi_1 - 1) z_{1t} + \left[B_{2n}\varphi_2 - (\lambda_1^2/2 + \lambda_2^2/2)\right] z_{2t}, \\ & \quad \left[(\lambda_1 + B_{1n})^2 + (\lambda_2 + \sigma_2 B_{2n})^2\right] z_{2t}\big). \end{aligned}$$

This implies a conditional cgf,

$$\begin{aligned} k_{n+1t}(s) &= \log E\left[e^{s(\log m_{t+1} + \log q_{t+1}^{(n)})} \mid z_{1t}, z_{2t}\right] \\ &= s\left[A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 + (B_{1n}\varphi_1 - 1) z_{1t} + \left[B_{2n}\varphi_2 - (\lambda_1^2/2 + \lambda_2^2/2)\right] z_{2t}\right] \\ &\quad + (s^2/2)\left[(\lambda_1 + B_{1n})^2 + (\lambda_2 + \sigma_2 B_{2n})^2\right] z_{2t}, \end{aligned}$$

which we then use to find the price of an $n+1$-maturity bond,

$$\begin{aligned}
\log q_t^{(n+1)} &= \log E_t\left[e^{\log m_{t+1}+\log q_{t+1}^{(n)}}\right] \\
&= k_{n+1t}(1) \\
&= \underbrace{A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2}_{=A_{n+1}} + \underbrace{(B_{1n}\varphi_1 - 1)}_{=B_{1n+1}} z_{1t} \\
&\quad + \Big[\underbrace{B_{2n}\varphi_2 - (\lambda_1^2/2+\lambda_2^2/2) + (\lambda_1+B_{1n})^2/2 + (\lambda_2+\sigma_2 B_{2n})^2/2}_{=B_{2n+1}}\Big] z_{2t}.
\end{aligned}$$

Recursions. Line up terms to find the recursive solutions for the intercept and slopes of the log bond price:

$$\begin{aligned}
A_{n+1} &= A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 \\
B_{1n+1} &= -(1-B_{1n}\varphi_1) \quad \Rightarrow \quad B_{1n} \;=\; -(1-\varphi_1^n)/(1-\varphi_1) \\
B_{2n+1} &= B_{2n}\varphi_2 - (\lambda_1^2/2+\lambda_2^2/2) + (\lambda_1+B_{1n})^2/2 + (\lambda_2+\sigma_2 B_{2n})^2/2 \\
&= B_{2n}(\varphi_2+\lambda_2\sigma_2) + \lambda_1 B_{1n} + B_{1n}^2/2 + B_{2n}^2\sigma_2^2/2,
\end{aligned}$$

and once again we start these recursions at $A_0 = B_{10} = B_{20} = 0$.

This isn't really much more complicated that the 1-factor CIR model. In fact, it's a lot like simply adding two of those models together. That's not a coincidence, of course, the model was set up from the beginning to deliver that feature. And again, we won't try to solve these by hand. But just like the 1-factor model, it's easy to program, which is what we'll do.

Risk-neutral distribution. Once again, we look for the location of the price-of-risk paramters, λ_1 and λ_2, to see if they can be absorbed into the parameters of the stochastic processes for the two factors. The persistence of z_{1t} is unaffected, so it will be the same in the risk-neutral distribution. The persistence of z_{2t}, however, adjusts to become $\varphi_2^* = \varphi_2 + \lambda_2\sigma_2$, hence the intercept must also adjust so that $\delta_2^* = \delta_2(1-\varphi_2)/(1-\varphi_2^*)$.

$$\begin{aligned}
\log m_{t+1}^* &= -i_t \\
z_{1t+1} &= (1-\varphi_1)\delta_1^* + \varphi_1 z_{1t} + \lambda_1 z_{2t} + z_{2t}^{1/2} w_{1t+1} \\
z_{2t+1} &= (1-\varphi_2^*)\delta_2^* + \varphi_2^* z_{2t} + \sigma_2 z_{2t}^{1/2} w_{2t+1}.
\end{aligned}$$

An interesting feature of this risk-neutral distribution is that z_{2t} affects the conditional *mean* of z_{1t+1} as well as its conditional variance.

Risk premium. The one-period-ahead forward rate is

$$f_t^{(1)} = (1-\varphi_1)\delta_1 + \varphi_1 i_t + (\lambda_1 - 1/2) z_{2t},$$

and the expected short rate is

$$E_t(i_{t+1}) \quad = \quad (1-\varphi_1)\delta_1 + \varphi_1 i_t,$$

so the risk premium is

$$f_t^{(1)} - E_t(i_{t+1}) \quad = \quad (\lambda_1 - 1/2)z_{2t},$$

which will vary as the conditional variance state variable, z_{2t}, varies. The unconditional mean of this risk premium is

$$\begin{aligned} E\left[f_t^{(1)} - E_t(i_{t+1})\right] &= (\lambda_1 - 1/2)E(z_{2t}) \\ &= (\lambda_1 - 1/2)\delta_2. \end{aligned}$$

(With a bit more effort, we could also show that the risk premiums on longer-maturity forward rates are also proportional to z_{2t}.)

Estimation. Now let's estimate parameters from the properties of forward rates reported in Table 3. This is a bit trickier than before, both because we have more parameters than before, but also because we will need to solve for things simultaneously.

Parameter values.

Let's try to estimate/calibrate the parameters of the stochastic processes for z_{1t} and z_{2t} first, without saying anything about their price of risk parameters, λ_1 and λ_2. Then solve for those from the shape of the average forward curve and it's volatility. Fortunately, the mean, variance, and autocorrelation of i_t will all be associated with just one factor, z_{1t}, so we can immediately choose values for the parameters for δ_1, φ_1, and δ_2, that worked in the previous model:

- *Mean.* We set δ_1 equal to the mean short rate: $\delta_1 = 5.1149/400 = 0.0128$.
- *Autocorrelation.* We set φ_1 equal to the observed autocorrelation of the short rate in the data, just as we did before. From the table, that gives us $\varphi_1 = 0.9699$.
- *Variance.* Similarly, the variance of the short rate is the variance of z_{1t}, which is $\delta_2/(1-\varphi_1^2)$. Given values for φ_1 and the variance of $(3.2874/400)^2$ in the data, we have $\delta_2 = 0.000004$.
- *Average forward premium.* We have 4 more parameters to identify $(\varphi_2, \sigma, \lambda_1, \lambda_2)$. Note that all 4 of these parameters affect forward rates of all maturities in a nonlinear fashion. We use a nonlinear solver to choose values for these parameters that make the mean forward premiums predicted by the model as close as possible to those in the data. The result is $\varphi_2 = 0.9999$, $\sigma = 0.000037$, $\lambda_1 = 77.7838$, and $\lambda_2 = -281.9028$.

When we plot the average forward premium from the data along with the prediction of the 2-factor CIR model that we just calibrated, it's difficult to tell the two curves apart. (The values in the data are the solid blue line and the model versions are the red asterisks.)

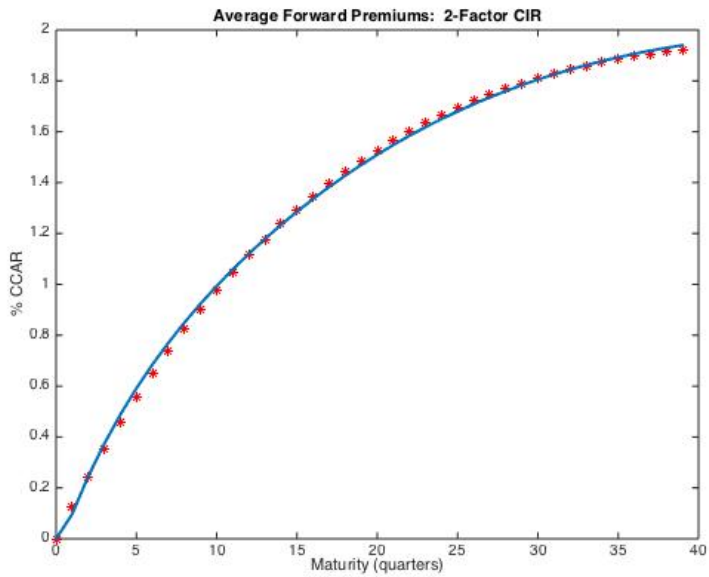

Figure 13: Comparison of predictions of the mean forward-rate curve implied by a calibration of the 2-factor CIR model to average forward rates in a sample from 1962-2016. Parameter values are $\varphi_1 = 0.9699$, $\varphi_2 = 0.9999$, $\delta_1 = 0.128$, $\delta_2 = 0.000004$, $\sigma = 0.000037$, $\lambda_1 = 77.7838$, and $\lambda_2 = -281.9028$.

There are a couple of features of this model worth noting. The stochastic process driving the short rate, z_{1t}, is highly persistent, $\varphi_1 = 0.9699$, but the stochastic process driving its conditional variance, z_{2t}, is even more persistent, $\varphi_2 = 0.9999$. To put that in context, the dependence of z_{1t+40} on z_{1t} is $\varphi_1^{40} = 0.2945$. The dependence of z_{2t+40} on z_{2t} is $\varphi_2^{40} = 0.9960$. In other words, the volatility factor is playing a major role in determining the riskiness of long-term bonds, while interest-rate risk is playing a much smaller role. Note also that the price of volatility risk is negative: a positive shock to volatility, w_{2t}, makes the pricing kernel go down.

Lastly, we can look at the standard deviations of forward rates implied by this calibrated 2-factor CIR model.

Long forward rates are perhaps a little too volatile in this calibration. But clearly we've succeeded in creating a term structure that more closely mimics the data on the dimensions of greatest interest.

We could continue this logic a little further and add a third factor (and maybe even a fourth?). From what we've seen, the performance

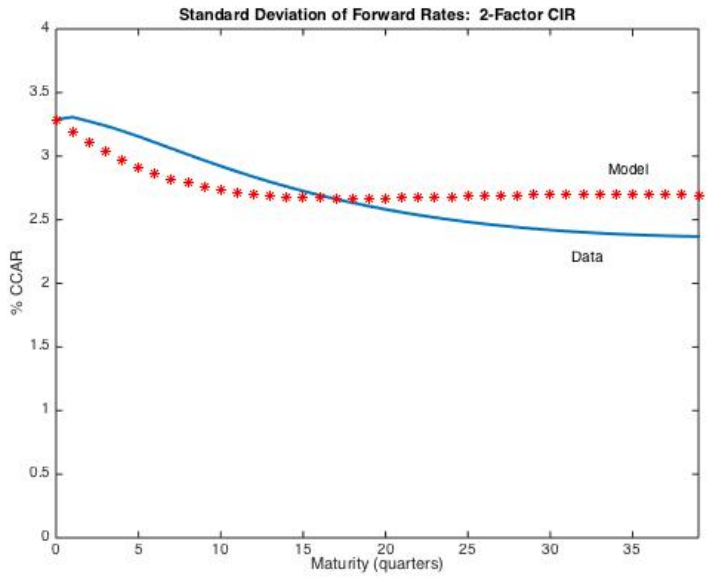

Figure 14: Comparison of predictions of the standard deviation of the forward-rate curve implied by a calibration of the 2-factor CIR model to the standard deviation of forward rates in a sample from 1962-2016. Parameter values are $\varphi_1 = 0.9699$, $\varphi_2 = 0.9999$, $\delta_1 = 0.128$, $\delta_2 = 0.000004$, $\sigma = 0.000037$, $\lambda_1 = 77.7838$, and $\lambda_2 = -281.9028$.

of the model will likely improve. The algebra is no more difficult, but the estimation of the additional parameters will become a much bigger bigger challenge. We could do it, but we would have to get far more serious about the statistical estimation methods we use, which would make this a course in advanced econometrics rather than asset pricing.

Digression: the "essentially affine" hack

There is another way to get a risk premium to depend on the state while maintaining a constant variance for the pricing kernel. It is nothing more than a *hack*, *i.e.*, a special trick that delivers a simple result but is difficult to interpret and does not generalize.

Let's go back to the original one-state-variable Vasicek model we started with summarized by two equations:

$$\begin{aligned} \log m_{t+1} &= -\lambda^2/2 + z_t + \lambda w_{t+1} \\ z_{t+1} &= (1-\varphi)\delta + \varphi z_t + \sigma w_{t+1}, \end{aligned}$$

but rather than changing both the λw_{t+1} and σw_{t+1} terms by inserting a $z_t^{1/2}$ as we did with the CIR model, let's only change the λw_{t+1} term in the first equation. Specifically, let's change the model to:

$$\begin{aligned} \log m_{t+1} &= -\lambda^2/2 - z_t + (\lambda z_t) w_{t+1} \\ z_{t+1} &= (1-\varphi)\delta + \varphi z_t + \sigma w_{t+1}, \end{aligned}$$

In other words, rather than changing the conditional standard deviation of w_{t+1} to make it proportional to $z_t^{1/2}$, let's leave it constant. Instead let's change the *price of risk* from a constant and make it proportional to z_t, *i.e.* λz_t.

OK... sure... why not? Well one reason is that we will no longer have the convenient log-linear solutions like we had in the Vasicek and CIR models. To see this, just try to calculate a one-period bond price with this model:

$$\begin{aligned} q_t^{(1)} = E_t[m_{t+1}] &= E_t[e^{-\lambda^2/2 - z_t + (\lambda z_t) w_{t+1}}] \\ &= e^{-\lambda^2/2 - z_t} E_t[e^{(\lambda z_t) w_{t+1}}] \\ &= e^{-\lambda^2/2 - z_t + \lambda^2 z_t^2/2} \\ \Rightarrow \log q_t^{(1)} &= -\lambda^2/2 - z_t + \lambda^2 z_t^2/2. \end{aligned}$$

This implies that the short rate will be a quadratic function of z_t. That doesn't seem like the end of the world. Quadratic functions aren't that complicated, are they?

Let's continue the exercise and use this new model to price a 2-period bond in the usual recursive way:

$$\begin{aligned} q_t^{(2)} = E_t[m_{t+1} q_{t+1}^{(1)}] &= E_t[e^{\log m_{t+1} + \log q_{t+1}^{(1)}}] \\ &= E_t[e^{-\lambda^2/2 - z_t + (\lambda z_t) w_{t+1} - \lambda^2/2 - z_{t+1} + \lambda^2 z_{t+1}^2/2}]. \end{aligned}$$

Uh oh!!! That expected value is no longer just evaluating the cumulant generating function for a normal random variable. What's the cumulant generating function for a quadratic function of a normal? We could probably figure that out (maybe it's related to a "Gamma" distribution?), but one thing for sure is that the cumulant generating function won't be a quadratic in z_t. Then when we go to price the 3-period bond, things will get even crazier. What a mess! The special feature we were able to exploit when solving the Vasicek and CIR models wasn't just that bond prices satisfied a recursive equation. The special feature was that recursive equation itself had a loglinear feature which this model doesn't deliver.

Here's the "essentially affine" hack that is quite popular in practice. Let me make a couple of simple changes to pricing kernel we started with – you'll have to look closely to even notice the difference. Our new model is

$$\begin{aligned} \log m_{t+1} &= -\lambda_t^2/2 - z_t + \lambda_t w_{t+1} \\ z_{t+1} &= (1-\varphi)\delta + \varphi z_t + \sigma w_{t+1}, \end{aligned}$$

where $\lambda_t = \lambda_0 + \lambda_1 z_t$. Do you see the difference? We made the price of risk a linear function of z_t rather than simply proportional. But that's not the "hack". We snuck another term that depends on z_t into the conditional mean of the pricing kernel: $-\lambda_t^2/2 = -(\lambda_0 + \lambda_1 z_t)^2/2$. Can you guess why? If you guessed "it's just a more convenient parameterization like before" or you guessed "because we have a fundamental economic theory that leads us to this functional form" you would be laughably wrong. It's just a hack! It's motivated only by the practical need to cancel out the term that is quadratic in z_t to maintain the loglinear structure.

Let's see how things differ with this hack in place. Given what we've seen before, we know what the conditional cdfs will be:

$$\begin{aligned} k_{zt}(s) &= s\big[(1-\varphi)\delta + \varphi z_t\big] + (s^2/2)\sigma^2 \\ k_{mt}(s) &= s(-\lambda_t^2/2 - z_t) + (s^2/2)\lambda_t^2. \end{aligned}$$

The short rate follows immediately:

$$i_t = y_t^{(1)} = -k_{mt}(1) = z_t.$$

We're back to the original Vasicek model with z_t scaled so that it's equal to the short rate. The hack seems to have simplified things tremendously.

Where was the magic trick? By hard-wiring the term $-\lambda_t^2/2$, which is a quadratic function of z_t, into the conditional mean of $\log m_{t+1}$, we created a perfect negative correlation between the conditional mean and the conditional variance of $\log m_{t+1}$. Recall the conditional variance of $\log m_{t+1}$ is $+\lambda_t^2$. When the conditional variance goes up, the conditional mean goes down by exactly the same amount, and vice versa. And we've rigged that in just the right way so that these two effect perfectly offset each other in the bond-pricing formula.

What about the $n+1$-maturity bond? Begin with the usual guess

$$\log q_t^{(n)} = A_n + B_n z_t,$$

and use this to solve for the sum of $\log m_{t+1}$ and $\log q_{t+1}^{(n)}$ in the usual way,

$$\log m_{t+1} + \log q_{t+1}^{(n)} = -\lambda_t^2/2 + A_n + B_n(1-\varphi)\delta + (B_n\varphi - 1)z_t + (\lambda_t + B_n\sigma)w_{t+1},$$

which implies a conditional cgf,

$$k_{n+1t}(s) = s\big[-\lambda_t^2/2 + A_n + B_n(1-\varphi)\delta + (B_n\varphi - 1)z_t\big] + (s^2/2)(\lambda_t + B_n\sigma)^2.$$

Use this to solve for the $n+1$-maturity bond price,

$$\begin{aligned}
\log q_t^{(n+1)} &= k_{n+1t}(1) \\
&= -\lambda_t^2/2 + A_n + B_n(1-\varphi)\delta + (B_n\varphi - 1)z_t + (\lambda_t + B_n\sigma)^2/2 \\
&= A_n + B_n(1-\varphi)\delta + (B_n\varphi - 1)z_t + B_n\sigma(\lambda_0 + \lambda_1 z_t) + B_n\sigma^2/2 \\
&= \underbrace{A_n + B_n[(1-\varphi)\delta + \lambda_0\sigma] + B_n^2\sigma^2/2}_{=A_{n+1}} + [\underbrace{B_n(\varphi + \lambda_1\sigma) - 1}_{=B_{n+1}}]z_t.
\end{aligned}$$

Recursions. Line up the coefficients for the log bond price and solve the recursive equations

$$\begin{aligned}
B_{n+1} &= B_n(\varphi + \sigma\lambda_1) - 1 \\
A_{n+1} &= A_n + B_n[(1-\varphi)\delta + \sigma\lambda_0] + B_n^2\sigma^2/2,
\end{aligned}$$

starting from initial conditions, $A_0 = 0$ and $B_0 = 0$.

Recall the risk-neutral distribution we calculated for the Vasicek model. It had the same persistence and volatility parameters, φ and σ as the original model, but it had a different intercept for z_t labeled δ^* that encoded the price of risk, λ. What is the comparable risk-neutral model in this "essentially affine" case?

Consider a risk-neutral model given by

$$\begin{aligned}
z_{t+1} &= (1-\varphi^*)\delta^* + \varphi^* z_t + \sigma w_{t+1} \\
\log m_{t+1}^* &= -i_t\ ,
\end{aligned}$$

where we haven't changed the conditional variance parameter the, σ, but we have changed both the persistence, φ^* and the mean δ^*, so that $\varphi^* = \varphi + \sigma\lambda_1$ and $(1-\varphi^*)\delta^* = (1-\varphi)\delta + \sigma\lambda_0$. Check the formulas for A_n and B_n. With these adjustments, they are exactly the same, so the two models will be indistinguishable.

We can see now why this hack has become so popular. The persistence of the of the short rate which we identified from data to be 0.97, is *not* the same as the persistent that governs risk premiums. Recall in the 2-factor CIR model we need that number to be 0.9999 which seemed pretty extreme (recall that number has to be smaller than 1, so we were taking a pretty thin slice off of the boundary condition). This model would seem to relieve that tension. But the cost is that we have no idea how to interpret that quadratic term in the log pricing kernel. It plays a magical role in solving the model, but it is exceptionally difficult to rationalize through economic fundamentals: what economic force would lead to conditional means and

conditional variances of the pricing kernel be perfectly negatively correlated?

Risk premiums. Start with the 1-period forward rate:

$$f_t^{(1)} - E_t(i_{t+1}) = (1-\varphi^*)\delta^* - (1-\varphi)\delta + (\varphi^* - \varphi)z_t.$$

Therefore, as long as the price of risk is not constant so that $\varphi^* \neq \varphi$, this model will have a time-varying risk premium. For an arbitrary maturity we have

$$f_t^{(n)} - E_t(i_{t+n}) = (1-\varphi^{*n})\delta^* - (1-\varphi^n)\delta + (\varphi^{*n} - \varphi^n)z_t.$$

Multi-factor essentially affine models. Adding factors to this model is straightforward. Consider a 2-factor example:[34]

$$\begin{aligned} z_{1t+1} &= (1-\varphi_1)\delta_1 + \varphi_1 z_{1t} + \sigma_1 w_{1t+1} \\ z_{2t+1} &= (1-\varphi_2)\delta_2 + \varphi_2 z_{2t} + \sigma_2 w_{2t+1} \\ \log m_{t+1} &= -\lambda_{1t}^2/2 - \lambda_{2t}^2/2 - z_{1t} - z_{2t} + \lambda_{1t} w_{1t+1} + \lambda_{2t} w_{2t+1} \end{aligned}$$

[34] Again, we could make this more complicated by building some correlation across the w's, or making the conditional mean of each z depend on the lagged value of the other, or making the λ's depend on both z's. Same algebra... many more parameters.

where $\lambda_{1t} = \lambda_{01} + \lambda_{11} z_{1t}$ and $\lambda_{2t} = \lambda_{02} + \lambda_{12} z_{2t}$. We assume, as before, that w_{1t} and w_{2t} are *iid* normals.

We would follow the same method as for the 2-factor Vasicek model and assume a loglinear structure for yields:

$$\log q_t^{(n)} = A_n + B_{1n} z_{1t} + B_{2n} z_{2t}$$

and use our familiar tricks to find A_n, B_{1n}, and B_{2n} as recursive functions of n and the other parameters of the model. We'll leave that as an exercise.

What about the risk-neutral version of this model? I think you can probably guess:

$$\begin{aligned} z_{1t+1} &= (1-\varphi_1^*)\delta_1^* + \varphi_1^* z_{1t} + \sigma_1 w_{1t+1} \\ z_{2t+1} &= (1-\varphi_2^*)\delta_2^* + \varphi_2^* z_{2t} + \sigma_2 w_{2t+1} \\ \log m_{t+1} &= -i_t, \end{aligned}$$

where σ_1 and σ_2 are unchanged, but we've cleverly chosen the other parameters to encode the price-of-risk parameters:

$$\begin{aligned} \varphi_1^* &= \varphi_1 + \sigma_1 \lambda_{11} \\ \varphi_2^* &= \varphi_2 + \sigma_2 \lambda_{12} \\ (1-\varphi_1^*)\delta_1^* &= (1-\varphi_1)\delta_1 + \sigma_1 \lambda_{01} \\ (1-\varphi_2^*)\delta_2^* &= (1-\varphi_2)\delta_2 + \sigma_2 \lambda_{02}. \end{aligned}$$

Empirical performance. Generally speaking, a multi-factor "essentially-affine" model will fit the data as well or better than a multi-factor CIR. Given it's simplicity and this empirical success, it's no surprise that this "hack" has been widely adopted by practitioners.

Asset-pricing theories

At this point, you may be asking yourself how the stochastic processes for the pricing kernel from a model like Vasicek or CIR, that we've identified from data on multi-period bond prices, relate to those in representative-agent structural models. Let's take a quick look.

Real vs. Nominal. Consider the expected utility pricing kernel with constant discounting and constant relative risk aversion:

$$\log m_{t+1} \quad = \quad \log \beta - \alpha g_{t+1},$$

where $g_{t+1} = \log(c_{t+1}/c_t)$ measures aggregate consumption growth. Note that this is a "real" pricing kernel – implicitly, the future payoffs that it discounts are denominated in units of consumption. But the data we've been looking at is for "nominal" bond prices. In other words, we need to make an inflation adjustment to this pricing kernel before we can price nominal bonds – or any other assets whose future payoffs are measured in dollars rather than the consumption those dollars purchase.

The logic is straightforward: imagine you had to pay $q_t^\$$ to purchase an asset with dollars, which we call the asset's *nominal* price.[35] If $p_t^\$$ is the dollar price of goods, *i.e.* the number of dollars I have to give up to get a unit of consumption, then trading $q_t^\$$ dollars for the asset is equivalent to trading $q_t^\$/p_t^\$$ units of consumption goods for the asset. We call this the *real* price of the asset. In the future the asset pays $d_{t+1}^\$$ in dollars. This is equivalent to the asset paying $d_{t+1}^\$/p_{t+1}^\$$ of consumption goods.

[35] Until now our notation for nominal and real assets has has been ambiguous. Here we need to be clear about that distinction so we will clutter up the notation by adding a superscript $ to all nominal variables. Real variables will be those *without* the $ subscript.

After dividing dollar-valued asset prices and dollar-valued payoffs by the appropriate dollar price of goods, all flows are now real-valued so we can apply the real pricing kernel in the usual way:

$$\begin{aligned}
(q_t^\$/p_t^\$) &= E_t\left[m_{t+1}(d_{t+1}^\$/p_{t+1}^\$)\right] \\
\Rightarrow q_t^\$ &= E_t\left[(m_{t+1}(p_t^\$/p_{t+1}^\$)d_{t+1}^\$\right] \\
&= E_t\left[m_{t+1}^\$ d_{t+1}^\$\right],
\end{aligned}$$

where we have implicitly defined the *nominal* pricing kernel,

$$\begin{aligned} m^{\$}_{t+1} &= m_{t+1}(p^{\$}_t / p^{\$}_{t+1}) \\ \Rightarrow \log m^{\$}_{t+1} &= \log m_{t+1} - \log(p^{\$}_{t+1}/p^{\$}_t) \end{aligned}$$

In any arbitrage-free environment, the nominal pricing kernel is the real pricing kernel *divided by* the growth in the dollar price of goods. Alternatively, the log of the nominal pricing kernel is the log of the real pricing kernel *minus* that rate of growth of the dollar price of goods, $\log(p^{\$}_{t+1}/p^{\$}_t)$, which we commonly call the *inflation rate*.

In the model in which consumption growth drives the real pricing kernel we have

$$\log m^{\$}_{t+1} = \log \beta - \alpha g_{t+1} - \log(p^{\$}_{t+1}/p^{\$}_t).$$

Let's set aside inflation – we'll get back to it shortly – and focus on the real pricing kernel. Note that it depends only on the stochastic process for consumption growth and two additional constant parameters, $0 < \beta < 1$ that models impatience, and $\alpha > 0$ that models risk aversion. The dynamics of the real pricing kernel in this model are driven entirely by the dynamics of consumption growth, g_{t+1}. What kind of dynamics would be consistent with observed bond prices? Since we're now talking about bonds that payoff in units of consumption rather than dollars, we need to first look at some data to get a better idea of what we're looking for.

Aside: TIPS bonds. The US Treasury issues bonds that are indexed to the rate of price inflation. That is, rather than paying a face value of \$1,000 in, say, 1 year, a Treasury Inflation-Protected Security (TIPS) bond would pay \$1,100 if the rate of inflation over the year is 10% or \$1,030 if that rate of inflation is 3%, *etc.* That is, it pays the equivalent purchasing power of \$1,000 today, which is like paying in units of aggregate consumption.[36] There are a smaller number of maturities issued – 5, 10, and 30 years – and the market only began in 1997. It also took a couple of years for the markets for these bonds to have enough trading volume and liquidity to generate meaningful prices. So we only have about 25 years of reliable data. Nonetheless, it's the best look we have at *real* bond prices.

Figure 15 plots the means and standard deviations for *real* forward rates implied by TIPS prices. The level is lower than we saw for comparable nominal bonds – recall that nominal bonds must provide added compensation for expected inflation that TIPS do not – but the average forward curve is still upward sloping and it's still highly volatile even 39-quarters out into the future. Moreover, the 1-year

[36] TIPS are also coupon bonds with their semi-annual coupons also indexed to measured inflation. Pure-discount bonds can be backed out of these coupon prices in the usual way. But there is an added complication associated with knowing which inflation rate applies to which payments. As a result, short-maturity – less than 1 year – prices are tricky. Firstly, because they are necessarily inferred from secondary-market trading of very old bonds, *i.e.*, those issued more than 4 years ago at the earliest and perhaps as long as 29 years ago. As a result, liquidity can be thin and prices noisy as a result. Also, there are complications with inflation measurement at high frequencies. We will, therefore, focus on properties of TIPS prices for maturities of 1-year and greater.

forward rate is highly persistent with a 1st-order autocorrelation of 0.91. Generally speaking, the properties of *real* bond yields are not all that different from *nominal* yields.

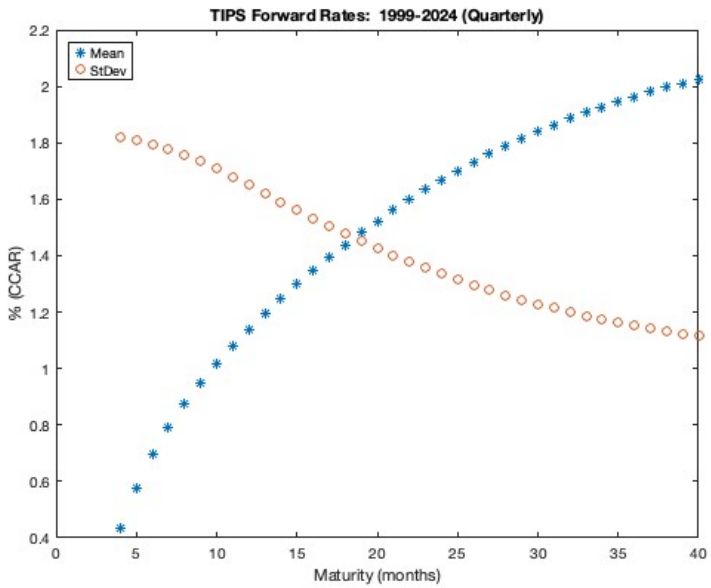

Figure 15: Comparison of predictions of the mean forward-rate curve implied by a calibration of the 2-factor CIR model to average forward rates in a sample from 1962-2016. Parameter values are $\varphi_1 = 0.9699$, $\varphi_2 = 0.9999$, $\delta_1 = 0.128$, $\delta_2 = 0.000004$, $\sigma = 0.000037$, $\lambda_1 = 77.7838$, and $\lambda_2 = -281.9028$.

What does all of this tell us about the real pricing kernel? We understand from our previous analysis that the dynamics of the real pricing kernel will determine both the persistence of real yields as well as the slope of the both the average real yield curve and it's volatility curve. And we now know that the dynamics of the real pricing kernel are due solely to the dynamics of aggregate consumption growth, so we need to explore how these thing line up in the data. We'll do this by looking at two standard models of consumption growth.

AR(1) consumption growth. Empirical macroeconomists often model consumption growth as an AR(1). Data on aggregate consumption growth suggests that there isn't much persistence in this process: reasonable estimates for the first-order autocorrelation coefficient might range from -0.3 to $+0.3$. This may not trouble macroeconomists, but for asset-pricing applications, an AR(1) without much persistence is highly problematic. We saw that the persistence of the real interest rate is the same as the persistence of the conditional mean of the pricing kernel. If consumption growth isn't very persistent, then the expected utility model with AR(1) consumption growth would predict that the real interest rate isn't very persistent either. But the data strongly suggest otherwise: the real interest rate as measured

through TIPS markets is *highly* persistent. This suggests that either the model of the pricing kernel or the model of consumption growth isn't very helpful... or perhaps both.

This suggests that either the model of the pricing kernel or the model of consumption growth isn't very helpful... or perhaps both. Let's keep the pricing kernel model but see if we can improve its predictions by making a simple change to the consumption-growth process. It would seem that what we need is a *wedge* between the persistence in the conditional mean which drives the highly persistent interest-rate process, and the persistence in the growth rate of consumption itself. The ARMA(1,1) model is the minimal extension of the AR(1) model – it adds just one parameter – that does exactly that.

ARMA(1,1) consumption growth. Consider extending the dynamic probability model for consumption growth to include a moving-average term along with the auto-regressive term – the so-called ARMA(1,1) model:

$$g_{t+1} = (1-\varphi)\mu_g + \varphi g_t + \sigma_g w_{t+1} - (\varphi - \sigma_x/\sigma_g)\sigma_g w_t.$$

This looks a bit odd until we realize that the parameters are chosen to reflect an AR(1) model for the conditional expected value,

$$\begin{aligned} g_{t+1} &= x_t + \sigma_g w_{t+1} \\ x_t &= (1-\varphi)\mu_g + \varphi x_{t-1} + \sigma_x w_t. \end{aligned}$$

A typical calibration matches the observed behavior of aggregate consumption (taken from Bansal-Yaron, *Journal of Finance*, 2004): $\mu_g = 0.0015$, $\varphi = 0.98$, $\sigma_g = 0.01$, and $\sigma_x = 0.001$. Note that even though the autocorrelation of the conditional mean in this calibration is 0.98, the autocorrelations of consumption growth is only 0.28, comparable to what we see in the data.

The log pricing kernel, therefore, is also an ARMA(1,1):

$$\log m_{t+1} = (1-\varphi)(\log\beta - \alpha\mu_g) + \varphi \log m_t + \sigma_g \alpha w_{t+1} - (\alpha\varphi\sigma - \alpha\sigma_g)w_t.$$

Recall that the pricing kernel in the Vasicek model was also an ARMA(1,1). On the surface, this would seem to suggest that the expected-utility pricing-kernel model with ARMA(1,1) consumption growth should match observed features of real bond pricing data at least as well as the Vasicek model. The question becomes can we choose values for α and β, the only parameters that are not constrained by the behavior of consumption growth, that result in a pricing kernel that will match observed *real* bond yields? Let's try.

The value of β only affects the mean of yields, so lets assume that we can always choose its value to match the mean of the real interest rate. We know that the value of φ of 0.98 in the Bansal-Yaron calibration will dictate the persistence of real yields and forward rates. Our data suggest a persistence parameter like 0.91 which is qualitatively similar, so we'll consider that a win. But notice what these values now imply for the moving-average coefficicient: $\alpha\varphi\sigma_x - \alpha\sigma_g = 12.79 \times 0.98 \times 0.001 - 12.79 \times 0.01 = -0.1154$ which, given the upward-sloping real yield curve, has the wrong sign! Recall that it's the sign of the moving-average coefficient that determines the sign of the autocorrelations in an ARMA(1,1). In other words, the pricing kernel in the expected utility model will be *positively* autocorrelated and the average yield curve will necessarily slope down. Again, this looks nothing like what we see in real world data.

Note that choosing a different value for α – either larger or smaller – will not change the sign of this moving average coefficient. That depends on the sign of $\varphi\sigma_x - \sigma_g$, and recall that this is the moving-average coefficient for aggregate consumption growth. For a negatively autocorrelated pricing kernel, *i.e.*, an upward-sloping average yield curve, we need this number to be positive. But consumption-growth data suggest that σ_g is 10-times larger than σ. You might think that making $\varphi < 1/10$ would help on this dimension, but the model would then be grossly at odds with the persistence of the real interest rate and other real yields. Moreover, would make the volatility graph converge to zero even faster as maturity grows, again, nothing like what we see in data.

We conclude that the expected utility model is, therefore, incapable of simultaneously matching the properties of consumption growth and basic features of multiperiod real bond prices, yields, and forward rates.

Real vs. Nominal.

What if we simply ignored the TIPS data, afterall it's a relatively short sample of observations, so maybe features like an upward-sloping yield curve are just artifacts of sampling error. Would we still have a *nominal* pricing kernel that matched the features of *nominal* multiperiod bonds?

If we specify persistent stochastic processes for consumption growth and for inflation, then this model looks a lot like a 2-factor Vasicek model. Is that likely to match the data? Let's give it a try. Let's define the first factor to be consumption growth, $z_{1t} = \log(c_t/c_{t+1})$, and the

second factor to be inflation $z_{2t} = \log(p^{\$}_{t+1}/p^{\$}_t)$, and assume both of these are AR(1) processes:[37]

$$\begin{aligned} z_{1t+1} &= (1-\varphi_1)\delta_1 + \varphi_1 z_{1t} + \sigma_1 w_{1t+1} \\ z_{2t+1} &= (1-\varphi_2)\delta_2 + \varphi_2 z_{2t} + \sigma_2 w_{2t+1}. \end{aligned}$$

[37] Note that these are no longer *abstract* state variables. These are now well-defined observable *macroeconomic* variables. As a result, we don't have the same freedom to change their parameters for our analytical convenience.

Substitute into the formula for the nominal pricing kernel and rearrange terms:

$$\begin{aligned} \log m^{\$}_{t+1} &= \log\beta - \alpha\log(c_{t+1}/c_t) - \log(p^{\$}_{t+1}/p^{\$}_t) \\ &= \log\beta - \alpha z_{1t+1} - z_{2t+1} \\ &= \log\beta - \alpha(1-\varphi_1)\delta_1 - (1-\varphi_2)\delta_2 \\ &\quad -\alpha\varphi_1 z_{1t} - \varphi_2 z_{2t} - \alpha\sigma_1 w_{1t+1} - \sigma_2 w_{2t+1}. \end{aligned}$$

Recall that the good performance of the 2-factor CIR model relied on having a factor that was really *really* persistent – we estimated $\varphi_2 = 0.9999$. There are two separate issues: the first is that persistence of inflation is not anything like that, and the second is that the second factor in the CIR model was stochastic volatility, which bears little resemblence to this inflation correction.

In other words, the nominal pricing kernel that corresponds to the expected utility model is incapable of simultaneously matching the properties of consumption growth, inflation, and basic features of multiperiod nominal bond prices, yields, and forward rates.

Can these structural models be saved? Yes! But this requires a different specification of the representative agent's preferences – one that will have a channel for stochastic volatility to affect the real pricing kernel – which we will leave for another time. "Recursive Utility" is a natural generalization of expected utility that delivers exactly the priced volatility risk the data seem to demand.) Another dimension is *endogenous monetary policy*: inflation isn't just an exogenous stochastic process, but rather is a choice variable of the government that will be correlated with variable like consumption growth. The Taylor Rule is a natural way to model monetary policy that has great potential for helping us understand the factors that drive multiperiod bond prices. But again, we'll save that analysis for another time.

It will also help to expand our notion of risk to include things like disasters, which we turn to now.

Extension: Disasters

Poisson disaster risk

Let's go back to the 1-factor Vasicek model but let's add some disaster risk. Begin with a state variable that is a normal AR(1), but only when conditioned on a particular value, j, of a Poisson random variable:

$$z_{t+1} \mid z_t, j \quad \sim \quad \mathcal{N}\big((1-\varphi)\delta + \varphi z_t - \theta j \,,\, \sigma^2\big), \tag{9}$$

where θ is a constant parameter and $j = 0, 1, 2, \ldots$. Big values for j imply a sudden drop in the conditional expected value for the state variable when $\theta > 0$, and a sudden increase when $\theta < 0$. Notice that when an event $j > 0$ occurs, it will have a persistent effect on future values of z_t since $\varphi \neq 0$.

The marginal distribution for j is a Poisson with intensity parameter ω,

$$p(j) = e^{-\omega}\omega^j / j!,$$

where $\omega > 0$.

We didn't put a time subscript on the variable j since it is *iid* (knowing the value of j today doesn't help you predict the value of j in the future). But keep in mind how the timing works. The disaster we're concerned with in equation (10) doesn't happen until date $t+1$. When we say "conditional on j" what we're really saying is "if we know this period the nature of a disaster that will occur next period, then here's how the distribution for z_{t+1} will look". Of course we don't know today whether a disaster will occur, so we have to form an expected value over that added random element. We've seen this before. What we'll get is a weighted average of normal distributions with the Poisson probabilities acting as the weights – the so-called *Poisson mixture of normals*. The only difference here is that we're taking a mixture of distributions conditional on z_t, which summarizes the information we have at date t.

Recall that the *conditional* cumulant generating function[38] for this random variable – conditional on the value of z_t but not j – is given by

$$k_{zt}(s) = \log E[e^{s\,z_{t+1}} \mid z_t] = s[(1-\varphi)\delta + \varphi z_t] + s^2\sigma^2/2 + \omega(e^{-s\theta} - 1).$$

[38] Our previous work with Poisson mixtures of normals did not have any dynamics as we have here. That is, the random variable z had a distribution conditional on a Poisson outcome j given by $z \mid j \;\sim \mathcal{N}(\mu - \theta j, \sigma^2)$. We saw that the cumulant generating function for z was then $k_z(s) = s\mu + s^2\sigma^2/2 + \omega(e^{-s\theta} - 1)$. It's exactly the same result here, but we've just replace the mean μ with the *conditional* mean $(1-\varphi_1)\delta + \varphi z_t$.

The other piece we need, of course, is a model of the pricing kernel. Conditional on the value of j, we assume that the log pricing kernel takes the usual form for a 1-factor Vasicek model:

$$\log m_{t+1} \mid z_t, j \quad \sim \quad \mathcal{N}\big(-\lambda_1^2/2 - \omega(e^{-\lambda_2\theta} - 1) - z_t - \lambda_2\theta j \ , \ \lambda_1^2\big).$$

There are a couple of things going on here that we should pause and think about. First, we've introduced two different price-of-risk parameters. The price of risk for z_t is given by λ_1, and the price of risk for j is given by λ_2. We do this since investors may view disasters are being more or less costly than other risk. We also snuck in an extra term, $-\omega(e^{-\lambda_2\theta} - 1)$, into the mean. The reason for this will be obvious shortly – no pun intended. (The *short* rate... get it?)

The conditional cumulant generating function for the log pricing kernel – conditional on the value of z_t but not j – is given by

$$\begin{aligned} k_{mt}(s) &= \log E[e^{s \log m_{t+1}} \mid z_t] \\ &= s[-\lambda_1^2/2 - \omega(e^{-\lambda_2\theta} - 1) - z_t] + s^2\lambda_1^2/2 + \omega(e^{-s\lambda_2\theta} - 1). \end{aligned}$$

This formula is very helpful for bond pricing.

Note that the short rate is quite easy to solve:

$$i_t \ = \ y_t^{(1)} \ = \ -k_{mt}(1) \ = \ z_t,$$

so once again, we've chosen the scale for z_t just right so that we can interpret it as the short rate. But notice that the stochastic process for z_t is now more complicated. It embodies two sources of risk: the risk from the normal random variable, and the disaster risk from the Poisson arrival. The disaster risk implies that the distribution for the interest rate will now involve more than just a mean and variance. The higher-order cumulants like skewness and kurtosis can now play an important role.

What about bonds of longer maturity? Follow our customary method of first guessing a loglinear structure,

$$\log q_t^{(n)} = A_n + B_n z_t,$$

and solve the implied recursions[39]

$$\log q_t^{(n+1)} \quad = \quad \log E_t\big[e^{\log m_{t+1} + \log q_{t+1}^{(n)}}\big].$$

Note that the conditional distribution of $\log m_{t+1} + \log q_{t+1}^{(n)}$ conditional on both z_t *and* j, is just a sum of two normals, hence its distribution is also normal

$$\mathcal{N}\big(-\lambda_1^2/2 - \omega(e^{-\lambda_2\theta} - 1) + A_n + B_n(1-\varphi)\delta + (B_n\varphi - 1)z_t - (\lambda_2 + B_n)\theta j \ , \ (\lambda_1 + B_n\sigma)^2\big).$$

[39] This is only a function of the state variable z_t. We also know the value of j at date t, but since it is assumed to be *iid*, it's value at date t contains no useful information for forecasting anything in the future.

The cumulant generating function conditional only on z_t, but not j, is given by

$$\begin{aligned} k_{n+1t}(s) &= \log E[e^{s\,(\log m_{t+1}+\log q_{t+1}^{(n)}} \mid z_t] \\ &= s[-\lambda_1^2/2 - \omega(e^{-\lambda_2\theta}-1) + A_n + B_n(1-\varphi)\delta + (B_n\varphi - 1)z_t] \\ &\qquad +s^2(\lambda_1 + B_n\sigma)^2/2 + \omega(e^{-s(\lambda_2+B_n)\theta}-1). \end{aligned}$$

Therefore, the $(n+1)$-maturity bond price can be found by evaluating the cumulant generating function at $s = 1$,

$$\begin{aligned} \log q_t^{(n+1)} &= k_{n+1t}(1) \\ &= \underbrace{A_n - \lambda_1^2/2 - \omega(e^{-\lambda_2\theta}-1) + B_n(1-\varphi)\delta + (\lambda_1 + B_n\sigma)^2/2 + \omega(e^{-(\lambda_2+B_n)\theta}-1)}_{=A_{n+1}} \\ &\qquad +(\underbrace{B_n\varphi - 1}_{=B_{n+1}})z_t. \end{aligned}$$

Recursions. The recursive formulas are similar to what we've seen before

$$\begin{aligned} B_{n+1} &= B_n\varphi - 1 \;\Rightarrow\; B_n = -(1-\varphi^n)/(1-\varphi) \\ A_{n+1} &= A_n - \lambda_1^2/2 - \omega(e^{-\lambda_2\theta}-1) + B_n(1-\varphi)\delta + (\lambda_1 + B_n\sigma)^2/2 + \omega(e^{-(\lambda_2+B_n)\theta}-1) \\ &= A_n + B_n[(1-\varphi)\delta + \lambda_1\sigma] + B_n^2\sigma^2/2 - \omega e^{-\lambda_2\theta}[1 - e^{-B_n\theta}]. \end{aligned}$$

and, as always, we build up these recursions from the values at $n = 0$ that insure $q_t^{(0)} = 1$, that is, $A_0 = B_0 = 0$.

Inspecting these formulas, we can see that adding the disaster to this model does not change the 1-factor structure of bond prices. What happens to the disaster variable j? It's presence is clearly affecting the distribution for the interest rate – it's no longer a normal random variable so it will have risk coming from higher-order cumulants. And since longer-maturity bonds are linear functions of the interest rate, they will inherit this behavior. In addition, it affects the mean of bond yields and forward rates of all maturities through the coefficients A_n. So overall, this is a more general model that could capture some interesting priced risk coming from fat-tailed distributions.

Risk-neutral distribution. A risk-neutral model with a slightly different distribution for both z_t and j will exactly replicate these bond prices:

$$\begin{aligned} z_{t+1} \mid z_t, j &\sim \mathcal{N}\big((1-\varphi)\delta^* + \varphi z_t - \theta j\,,\ \sigma^2\big) \\ p^*(j) &= e^{-\omega^*}\omega^{*j}/j! \\ \log m_{t+1}^* &= -i_t, \end{aligned}$$

where $\delta^* = \delta + \lambda_1\sigma/(1-\varphi)$ and $\omega^* = \omega e^{-\lambda_2\theta}$. As in the Vasicek model, the persistence parameter, φ, and the conditional conditional variance parameter, σ, are the same, but the intercept parameters changes from δ to δ^*. In addition, the Poisson intensity parameter in the risk-neutral distribution changes from ω to ω^* to capture the price of disaster risk, λ_2.

But the one thing that does not change in this model is that, just like the original Vasicek model, risk premiums will be constant functions of maturity. (Recall: $B_1 = -1$ and $B_1 - B_2 = \varphi$.)

$$\begin{aligned} f_t^{(1)} &= \log q_t^{(1)} - \log q_t^{(2)} \\ &= (A_1 - A_2) + (B_1 - B_2)z_t \\ &= (1-\varphi)\delta + \lambda_1\sigma - \sigma^2/2 + \omega e^{-\lambda_2\theta}(1-e^{\theta}) + \varphi z_t \end{aligned}$$

$$E_t(i_{t+1}) = E_t(z_{t+1}) = [de^{k_{zt}(s)}/ds]|_{s=0} = (1-\varphi)\delta - \omega\theta + \varphi z_t$$

$$\Rightarrow f_t^{(1)} - E_t(i_{t+1}) = \lambda_1\sigma - \sigma^2/2 + \omega\theta + \omega e^{-\lambda_2\theta}(1-e^{\theta}).$$

Note that when $\omega = 0$ the risk premium is the same as the Vasicek model. But when $\omega \neq 0$, there is risk coming from the higher-order moments of z_t beyond the variance. But most importantly, this risk premium will *not* vary across time. And a similar result will hold for risk premiums on longer-maturity forward rates. Given what we know about the data, *e.g.*, the variance of the forward spread is large even for very long-maturity bonds, this is a serious limitation of this model.

Persistent Poisson disaster risk

What if we made a small change to this model by introducing an probability intensity parameter that was itself a persistent stochastic process? That is, assume that the distribution for the arrival of a disaster at date $t+1$ was given by the Poisson distribution

$$p_t(j_{t+1}) = e^{-\omega z_{2t}}(\omega z_{2t})^{j_{t+1}}/j_{t+1}!\ ,$$

for $j_{t+1} = 0, 1, 2, \ldots$. The notation is a little awkward – we had previously been suppressing the subscript $t+1$ on j, but now the timing will be important to track. The key feature we've added is a Poisson intensity that is no longer just a constant parameter, ω, but rather it a stochastic process ωz_{2t}, where the conditional distribution for z_{2t} is given by

$$z_{2t+1} \mid z_{2t} \sim \mathcal{N}\big((1-\varphi_2)\delta_2 + \varphi_2 z_{2t}\ ,\ \sigma_2^2\big).$$

Now the expected value of the intensity parameter $E[\omega z_{2t}] = \omega\delta_2$, but since the intensity parameter is changing through time in a persistent way, the Poisson probability of a disaster is changing through time as well. [40] In other words, not only can the probability change from period to period, but it is persistent: if the probability of a disaster is higher than average this period, it's is likely to continue to be higher than average next period too. As a result, a disaster event this period isn't guaranteed to recur next period, but it is highly likely.

[40] Recall that the intensity in a Poisson probability distribution must be a postive number. Modeling it as a normal random variable introduces the potential that it could take on negative values. Recall we faced a similar issue when modeling the conditional variance with a normal distribution in the 2-factor CIR model. We will assume for the time being that the paramter values we use for ω, φ_2 and σ_2 are carefully chosen so that the probability of a negative value is vanishingly small. In other words... we'll just ignore that unpleasant fact for the time being and perhaps come back to it later.

How does this change our bond-pricing model? Let's start over from the beginning. The distribution of the state variable z_{1t+1} conditional on a disaster j_{t+1} is similar to what we had before,

$$z_{1t+1} \mid z_{1t}, j_{t+1} \quad \sim \quad \mathcal{N}\big((1-\varphi_1)\delta_1 + \varphi_1 z_{1t} - \theta j_{t+1}\ ,\ \sigma_1^2\big).$$

We've added the subscript 1 to everything to keep things separate from the intensity parameters which have a subscript 2.

The conditional cumulant generating function for this random variable – conditional on the value of z_{1t} *and* z_{2t}, but *not* j_{t+1} – is given by

$$k_{z_{1t}}(s) = \log E[e^{s\, z_{1t+1}} \mid z_{1t}\ , z_{2t}] = s[(1-\varphi_1)\delta_1 + \varphi_1 z_{1t}] + s^2\sigma_1^2/2 + \omega z_{2t}(e^{-s\theta} - 1).$$

This is just what we had iid case, but now ωz_{2t} is playing the role of the constant value of ω we used before.

Let's model the pricing kernel in the same way as before. Conditional on the value of j_{t+1} as well as z_{1t} and z_{2t}, we assume that the log pricing kernel takes a form similar to a 2-factor Vasicek model:

$$\log m_{t+1} \mid z_{1t}\ , z_{2t}\ , j_{t+1} \quad \sim \quad \mathcal{N}\big(-\lambda_1^2/2 - z_{1t} - (e^{-\lambda_2\theta} - 1)\omega z_{2t} - \lambda_2\theta j_{t+1}\ ,\ \lambda_1^2\big).$$

The conditional cumulant generating function for the log pricing kernel – conditional on the value of z_{1t} and z_{2t} but not j_{t+1} – is given by

$$\begin{aligned} k_{mt}(s) &= \log E[e^{s \log m_{t+1}} \mid z_{1t}\ , z_{2t}] \\ &= s[-\lambda_1^2/2 - z_{1t} - (e^{-\lambda_2\theta} - 1)\omega z_{2t}] + s^2\lambda_1^2/2 + \omega z_{2t}(e^{-s\lambda_2\theta} - 1). \end{aligned}$$

From this formula we immediately find the short rate

$$\begin{aligned} i_t = y_t^{(1)} &= -\log E\big[e^{\log m_{t+1}} \mid z_{1t}\ , z_{2t}\big] \\ &= -k_{mt}(1) \ = \ z_{1t}. \end{aligned}$$

Note that we engineered the conditional mean of the pricing kernel so that we can interpret the first factor as the short rate. This isn't necessary, of course, but it does make the model easier to interpret, and is analogous to what we did in the 2-factor CIR model.

What about bonds of longer maturity? Once again, we follow our customary method of first guessing a loglinear structure,

$$\log q_t^{(n)} = A_n + B_{1n} z_{1t} + B_{2n} z_{2t},$$

and solve the implied recursions

$$\log q_t^{(n+1)} \quad = \quad \log E\big[e^{\log m_{t+1} + \log q_{t+1}^{(n)}} \mid z_{1t}\,,\, z_{2t}\big].$$

Note first that the conditional distribution of $\log m_{t+1} + \log q_{t+1}^{(n)}$ conditional on both state variables, z_{1t} and z_{2t}, *and* j_{t+1}, is still just a sum of two normals, hence its distribution is also normal. The conditional mean is

$$-\lambda_1^2/2 + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 + (B_{1n}\varphi_1 - 1)z_{1t} + [B_{2n}\varphi_2 - \omega(e^{-\lambda_2\theta} - 1)]z_{2t} - (\lambda_2 + B_{1n})\theta j_{t+1}$$

and the conditional variance is

$$(\lambda_1 + B_{1n}\sigma_1 + B_{2n}\sigma_2)^2.$$

There are more terms to keep track of, but that's just algebra. Conceptually, nothing is very new here.

The cumulant generating function of of this linear combination of normals conditional on z_{1t} and z_{2t} but *not* j_{t+1}, is given by

$$\begin{aligned} k_{n+1t}(s) &= \log E[e^{s\,(\log m_{t+1} + \log q_{t+1}^{(n)}} \mid z_{1t}\,,\, z_{2t}] \\ &= s\big(-\lambda_1^2/2 + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 + (B_{1n}\varphi_1 - 1)z_{1t} + [B_{2n}\varphi_2 - \omega(e^{-\lambda_2\theta} - 1)]z_{2t}\big) \\ &\quad + s^2(\lambda_1 + B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2 + \omega z_{2t}(e^{-s(\lambda_2 + B_{1n})\theta} - 1). \end{aligned}$$

Therefore, the $n+1$-maturity bond price can be found by evaluating the cumulant generating function at $s = 1$,

$$\begin{aligned} \log q_t^{(n+1)} &= k_{n+1t}(1) \\ &= -\lambda_1^2/2 + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 + (B_{1n}\varphi_1 - 1)z_{1t} + [B_{2n}\varphi_2 - \omega(e^{-\lambda_2\theta} - 1)]z_{2t} \\ &\quad + (\lambda_1 + B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2 + \omega z_{2t}(e^{-(\lambda_2 + B_{1n})\theta} - 1) \\ &= \underbrace{A_n + B_{1n}[(1-\varphi_1)\delta_1 + \lambda_1\sigma_1] + B_{2n}[(1-\varphi_2)\delta_2 + \lambda_1\sigma_2] + (B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2}_{A_{n+1}} \\ &\quad + \underbrace{(B_{1n}\varphi_1 - 1)}_{B_{1n+1}} z_{1t} + \underbrace{[B_{2n}\varphi_2 + \omega e^{-\lambda_2\theta}(e^{-B_{1n}\theta} - 1)]}_{B_{2n+1}} z_{2t} \end{aligned}$$

Recursions. The recursive formulas are similar to what we've seen before

$$\begin{aligned}
B_{1n+1} &= B_n\varphi_1 - 1 \;\Rightarrow\; B_{1n} = -(1-\varphi_1^n)/(1-\varphi_1) \\
B_{2n+1} &= B_{2n}\varphi_2 + \omega e^{-\lambda_2\theta}(e^{-B_{1n}\theta} - 1) \\
A_{n+1} &= A_n + B_{1n}[(1-\varphi_1)\delta_1 + \lambda_1\sigma_1] + B_{2n}[(1-\varphi_2)\delta_2 + \lambda_1\sigma_2] + (B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2.
\end{aligned}$$

and, as always, we build up these recursions from the values at $n = 0$ that insure $q_t^{(0)} = 1$, that is, $A_0 = B_0 = 0$. Notice that we solve these sequentially: B_{1n} depends only on φ_1, so solve this first; then given the value of B_{1n} and other parameters, we can solve B_{2n+1}; finally, given values for B_{1n} and B_{2n} and the other parameters, we can solve for A_{n+1}.

Risk-neutral distribution. Notice where the price-of-risk parameters, λ_1 and λ_2, appear in these recursive formulas. The terms involving the persistence paramters, φ_1 and φ_2, and the volatility parameters, σ_1 and σ_2, are unaffected by the prices of risk, so they will be the same for the risk-neutral distribution. If we replaced δ_1 with δ_1^* such that $\delta_1^* = (1-\varphi_1)\delta_1 + \lambda_1\sigma_1/(1-\varphi_1)$, and δ_2 with δ_2^* such that $\delta_2^* = \delta_2 + \lambda_1\sigma_2/(1-\varphi_2)$, and ω with ω^* such that $\omega^* = \omega e^{-\lambda_2\theta}$, then the following risk-neutral pricing kernel with those adjustments to the stochastic processes, will produce the same bond prices:

$$\begin{aligned}
z_{1t+1} \mid z_{1t}, j_{t+1} &\sim \mathcal{N}\big((1-\varphi_1)\delta_1^* + \varphi_1 z_t - \theta j_{t+1}\,,\, \sigma_1^2\big) \\
z_{2t+1} \mid z_{2t} &\sim \mathcal{N}\big((1-\varphi_2)\delta_2^* + \varphi_2 z_t\,,\, \sigma_2^2\big) \\
p_t^*(j_{t+1}) &= e^{-\omega^* z_{2t}}(\omega^* z_{2t})^{j_{t+1}}/j_{t+1}! \\
\log m_{t+1}^* &= -i_{1t}.
\end{aligned}$$

Finally, the similarity of this model and the 2-factor CIR model should be fairly obvious. And given that similarity, this model has the potential to fit the facts as well as that model did. The interpretation, however is quite different. Remember, z_{2t} is a Poisson intensity parameter that we're allowing to move through time in a persistent way. In the 2-factor CIR model, z_{2t} was a random variance that we were allowing to move through time in a persistent way.

Essentially-affine model with persistent disaster risk

State variable conditional on knowing the future disaster state:

$$z_{1t+1} \mid z_{1t}, j_{t+1} \sim \mathcal{N}\big((1-\varphi_1)\delta_1 + \varphi_1 z_{1t} - \theta j_{t+1}\,,\, \sigma_1^2\big).$$

Poisson probability with an AR(1) intensity:

$$p_t(j_{t+1}) = e^{-\omega z_{2t}}(\omega z_{2t})^{j_{t+1}}/j_{t+1}!\ ,$$

for $j = 0, 1, 2, \ldots$.

$$z_{2t+1} \mid z_{2t} \quad \sim \quad \mathcal{N}\big((1-\varphi_2)\delta_2 + \varphi_2 z_{2t}\ ,\ \sigma_2^2\big).$$

Cumulant generating function for z_{1t+1} conditional only on z_{1t} and z_{2t}, but not j_{t+1}:

$$k_{z_1t}(s) = \log E[e^{s\,z_{1t+1}} \mid z_{1t}\ ,\ z_{2t}] = s[(1-\varphi_1)\delta_1 + \varphi_1 z_{1t}] + s^2\sigma_1^2/2 + \omega z_{2t}(e^{-s\theta} - 1).$$

Essentially-affine pricing kernel conditional on knowing the future disaster state:

$$\log m_{t+1} \mid z_{1t}\ ,\ z_{2t}\ ,\ j_{t+1} \quad \sim \quad \mathcal{N}\big(-\lambda_{1t}^2/2 - z_{1t} - (e^{-\lambda_2\theta} - 1)\omega z_{2t} - \lambda_2\theta j_{t+1}\ ,\ \lambda_{1t}^2\big),$$

where $\lambda_{1t} = \lambda_0 + \lambda_1 z_{1t}$.

Cumulant generating function for the pricing kernel conditional only on z_{1t} and z_{2t}, but not j_{t+1}:

$$\begin{aligned} k_{mt}(s) &= \log E[e^{s\,\log m_{t+1}} \mid z_{1t}\ ,\ z_{2t}] \\ &= s[-\lambda_{1t}^2/2 - z_{1t} - (e^{-\lambda_2\theta} - 1)\omega z_{2t}] + s^2\lambda_{1t}^2/2 + \omega z_{2t}(e^{-s\lambda_2\theta} - 1). \end{aligned}$$

Short rate:

$$\begin{aligned} i_t \;=\; y_t^{(1)} &= -\log E\big[e^{\log m_{t+1}} \mid z_{1t}\ ,\ z_{2t}\big] \\ &= -k_{mt}(1) \;=\; z_{1t}. \end{aligned}$$

Cumulant generating function for the $n+1$-maturity bond:

$$\begin{aligned} k_{n+1t}(s) &= \log E[e^{s\,(\log m_{t+1} + \log q_{t+1}^{(n)}} \mid z_{1t}\ ,\ z_{2t}] \\ &= s\big(-\lambda_{1t}^2/2 + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 + (B_{1n}\varphi_1 - 1)z_{1t} + [B_{2n}\varphi_2 - \omega(e^{-\lambda_2\theta} - 1)]z_{2t}\big) \\ &\quad + s^2(\lambda_{1t} + B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2 + \omega z_{2t}(e^{-s(\lambda_2 + B_{1n})\theta} - 1). \end{aligned}$$

Price of $n+1$-maturity bond:

$$\begin{aligned} \log q_t^{(n+1)} &= k_{n+1t}(1) \\ &= -\lambda_{1t}^2/2 + A_n + B_{1n}(1-\varphi_1)\delta_1 + B_{2n}(1-\varphi_2)\delta_2 + (B_{1n}\varphi_1 - 1)z_{1t} + [B_{2n}\varphi_2 - \omega(e^{-\lambda_2\theta} - 1)]z_{2t} \\ &\quad + (\lambda_{1t} + B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2 + \omega z_{2t}(e^{-(\lambda_2 + B_{1n})\theta} - 1) \\ &= \underbrace{A_n + B_{1n}[(1-\varphi_1)\delta_1 + \lambda_0\sigma_1] + B_{2n}[(1-\varphi_2)\delta_2 + \lambda_0\sigma_2] + (B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2}_{A_{n+1}} \\ &\quad + \underbrace{[B_{1n}(\varphi_1 + \lambda_1\sigma_1) - 1]}_{B_{1n+1}} z_{1t} + \underbrace{[B_{2n}(\varphi_2 + \lambda_1\sigma_2) + \omega e^{-\lambda_2\theta}(e^{-B_{1n}\theta} - 1)]}_{B_{2n+1}} z_{2t} \end{aligned}$$

Recursions.

$$
\begin{aligned}
B_{1n+1} &= B_{1n}(\varphi_1 + \lambda_1\sigma_1) - 1 \\
B_{2n+1} &= B_{2n}(\varphi_2 + \lambda_1\sigma_2) + \omega e^{-\lambda_2\theta}(e^{-B_{1n}\theta} - 1) \\
A_{n+1} &= A_n + B_{1n}[(1-\varphi_1)\delta_1 + \lambda_0\sigma_1] + B_{2n}[(1-\varphi_2)\delta_2 + \lambda_0\sigma_2] + (B_{1n}\sigma_1 + B_{2n}\sigma_2)^2/2.
\end{aligned}
$$

Risk-neutral distribution.

$$
\begin{aligned}
z_{1t+1} \mid z_{1t}, j_{t+1} &\sim \mathcal{N}\big((1-\varphi_1^*)\delta_1^* + \varphi_1^* z_{1t} - \theta j_{t+1}\ ,\ \sigma_1^2\big) \\
z_{2t+1} \mid z_{2t} &\sim \mathcal{N}\big((1-\varphi_2^*)\delta_2^* + \varphi_2^* z_{2t}\ ,\ \sigma_2^2\big) \\
p_t^*(j_{t+1}) &= e^{-\omega^* z_{2t}}(\omega^* z_{2t})^{j_{t+1}}/j_{t+1}! \\
\log m_{t+1}^* &= -i_t,
\end{aligned}
$$

where

$$
\begin{aligned}
\varphi_1^* &= \varphi_1 + \lambda_1\sigma_1 \\
\varphi_2^* &= \varphi_2 + \lambda_1\sigma_2 \\
\omega^* &= \omega e^{-\lambda_2\theta} \\
\delta_1^* &= [(1-\varphi_1)\delta_1 + \lambda_0\sigma_1]/(1-\varphi_1^*) \\
\delta_2^* &= [(1-\varphi_2)\delta_2 + \lambda_0\sigma_2]/(1-\varphi_2^*).
\end{aligned}
$$

Bottom line

Bond pricing in Markov environments is a classic application of dynamic asset pricing theory. We get to use what we learned about stochastic processes to characterize bond prices of all maturities as functions of the current state. These prices follow, as always, from the no-arbitrage theorem. The Vasicek model has a particularly user-friendly structure, in which bond prices are loglinear functions of the state. The latter is special, but illustrates how the dynamics of the state, typically captured by the autoregressive parameter φ, are reflected in long-maturity bond prices. Mean reversion, for example, tends to reduce the sensitivity of long bond prices to the state. In this sense, the cross section of bond prices incorporates the time series behavior of interest rates.

We can use similar methods to price other assets. The new ingredient with other assets is cash flows, which themselves might be modeled as functions of the state.

Practice problems

1. *Bond basics.* Consider the bond prices at some date t:

Maturity n	Price $q_t^{(n)}$
0	1.0000
1 year	0.9512
2 years	0.8958
3 years	0.8353
4 years	0.7788
5 years	0.7261

(a) What are the yields $y_t^{(n)}$?

(b) What are the forward rates $f_t^{(n-1)}$?

(c) Suppose $f_t^{(1)}$ rises by 1 percent. How does $y_t^{(1)}$ change? How does $y_t^{(2)}$ change?

Answer.

Maturity n	Price $q_t^{(n)}$	Yield $y_t^{(n)}$	Forward $f_t^{(n-1)}$
0	1.0000		
1 year	0.9512	0.0500	0.0500
2 years	0.8958	0.0550	0.0600
3 years	0.8353	0.0600	0.0700
4 years	0.7788	0.0625	0.0700
5 years	0.7261	0.0640	0.0700

(a) See above.

(b) See above.

(c) If $f_t^{(1)}$ rises by 1%, then $y_t^{(1)} = f_t^{(0)}$ doesn't change and $y_t^{(2)} = (f_t^{(0)} + f_t^{(1)})/2$ rises by 0.5%.

2. *Moving average bond pricing.* Consider the bond pricing model

$$\begin{aligned} \log m_{t+1} &= -\lambda^2/2 - x_t + \lambda w_{t+1} \\ x_t &= \delta + \sigma(w_t + \theta w_{t-1}). \end{aligned}$$

Here x_t need not correspond to the state z_t.

(a) What is the short rate $f_t^{(0)}$?

(b) Suppose bond prices take the form

$$\log q_t^{(n)} \quad = \quad A_n + B_n w_t + C_n w_{t-1}.$$

Use the pricing relation (1) to derive recursions connecting $(A_{n+1}, B_{n+1}, C_{n+1})$ to (A_n, B_n, C_n). What are (A_n, B_n, C_n) for $n = 0, 1, 2, 3$?

(c) Express forward rates as functions of the state (w_t, w_{t-1}). What are f_t^1 and f_t^2?

(d) What is $E(f^{(1)} - f^{(0)})$? What parameters govern its sign?

Answer.

(a) The short rate is

$$f_t^{(0)} \quad = \quad -\log E_t m_{t+1} \quad = \quad \lambda^2/2 + x_t - \lambda^2/2 \quad = \quad x_t.$$

The second equality is the usual "mean plus variance over two" with the sign flipped (as indicated by the first equality). In other words: the usual setup. In what follows, we'll kill off x_t by substituting.

(b) Bond prices follow from the pricing relation,

$$q_t^{(n+1)} \quad = \quad E_t(m_{t+1} q_{t+1}^{(n)}),$$

starting with $n = 0$ and $q_t^0 = 1$. The state in this case is (w_t, w_{t-1}), a simple example of a two-dimensional model, hence the extra term in the form of the bond price. We need

$$\log(m_{t+1} q_{t+1}^{(n)}) \quad = \quad A_n - (\lambda^2/2 + \delta) + (\lambda + B_n) w_{t+1} + (C_n - \sigma) w_t - \sigma\theta w_{t-1}.$$

The (conditional) mean and variance are

$$\begin{aligned} E_t[\log(m_{t+1} q_{t+1}^{(n)})] &= A_n - (\lambda^2/2 + \delta) + (C_n - \sigma) w_t - \sigma\theta w_{t-1} \\ \mathrm{Var}_t[\log(m_{t+1} q_{t+1}^{(n)})] &= (\lambda + B_n)^2. \end{aligned}$$

Using "mean plus variance over two" and lining up terms gives us

$$\begin{aligned} A_{n+1} &= A_n - (\lambda^2/2 + \delta) + (\lambda + B_n)^2/2 \\ &= A_n - \delta + \lambda B_n + (B_n)^2/2 \\ B_{n+1} &= C_n - \sigma \\ C_{n+1} &= -\sigma\theta \end{aligned}$$

for $n = 0, 1, 2, \ldots$. That gives us

n	A_n	B_n	C_n
0	0	0	0
1	$-\delta$	$-\sigma$	$-\sigma\theta$
2	$-2\delta - \lambda\sigma + \sigma^2/2$	$-\sigma(1+\theta)$	$-\sigma\theta$
3	X	$-\sigma(1+\theta)$	$-\sigma\theta$

with $X = -3\delta - \lambda(2+\theta) + [1+(1+\theta)^2]\sigma^2/2$.

(c) In general, forward rates are

$$f_t^{(n)} = (A_n - A_{n+1}) + (B_n - B_{n+1})w_t + (C_n - C_{n+1})w_{t-1}.$$

That gives us

$$\begin{aligned} f_t^{(0)} &= \delta + \sigma w_t + \sigma\theta w_{t-1} \\ f_t^{(1)} &= \delta + \lambda\sigma - \sigma^2/2 + \sigma\theta w_t \\ f_t^{(2)} &= \delta - (1+\theta)^2\sigma^2/2 + \lambda\sigma(1+\theta). \end{aligned}$$

(d) The means are the same with $w_t = w_{t-1} = 0$, their mean. Therefore

$$E(f^{(1)} - f^{(0)}) = \lambda\sigma - \sigma^2/2.$$

Therefore we need $\lambda\sigma > \sigma^2/2$, so a necessary condition is that λ and σ have the same sign.

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